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DIPLOMARBEIT

Plant Up!

Smart Plant Monitoring with IoT and Gamification

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Wien, 7. März 2026

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Danksagungen

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Abstract

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Chapter 1

Introduction

The "Plant Boom" vs. The Reality

The indoor plant market experienced a massive rise during the early 2020s, with people spending more money on indoor plants. From $1.31 billion in 2019 to $2.17 billion by 2021 [1]. This "boom" saw household participation in indoor gardening reach a four-year high of 38.1 million households [2]. However, even with this rise in popularity, the reality is that people are not very good at taking care of plants. While demand rose by 18%, failure rates remained constant: approximately 40% of plants perish within the supply chain before reaching a shelf [3], and another 35% die shortly after entering a customer's home [4]. This indicates that the industry's economic "boom" is significantly fueled by a replacement cycle where consumers repeatedly purchase new plants to compensate for those lost.

What is "Plant Up!" ?

Plant Up! is a smart gardening system that aims to solve this problem by providing users with the tools and knowledge they need to keep their plants alive. It combines IoT sensors with a mobile application to create a seamless and user-friendly experience [5]. The system monitors soil moisture, light, and temperature in real-time, and provides users with actionable insights and guidance on how to care for their plants.

What "Plant Up!" offers

Plant Up! [5] offers a range of features designed to help users keep their plants alive. These include:

- Real-time monitoring of soil moisture, light, and temperature

- Actionable insights and guidance on how to care for plants
- A comprehensive database of plants and their care requirements
- A community forum for users to share tips and advice

Chapter 2

Combining Nature and Technology: How Sensor-Based and Data-Driven Technologies Enhance Plant Monitoring and Care

Author: Arun Sherzai

2.1 Introduction

Houseplants are a common feature in modern households, with the global indoor plant market valued at approximately 21 billion USD in 2025 [6]. Yet maintaining them over longer periods remains challenging for many people. One of the most common reasons is improper care rather than neglect. In particular, horticultural experts identify overwatering as the leading cause of houseplant failure, frequently resulting in yellowing leaves, wilting, and root rot [7]. This indicates a fundamental issue: plants often suffer because their actual needs are misunderstood.

A key reason for this misunderstanding lies in the limitations of human judgment. Soil moisture is commonly assessed by touching the surface of the soil, even though moisture levels can differ significantly between the upper layer and the root zone [7]. As a result, soil may appear dry while deeper layers remain saturated, which increases the risk of root diseases and root rot [8]. These processes occur below the surface and are therefore difficult to detect without technical support.

Human perception of light conditions is similarly unreliable. The visual system adapts to changing brightness levels, which often leads to an overestimation of the usable light available in indoor environments [9]. A room may appear bright to humans while still providing insufficient light for healthy plant growth.

On the other side, sensor technology and Internet of Things (IoT) applications are becoming increasingly widespread. The global smart home market is experiencing strong

growth, driven mainly by automation in areas such as lighting, climate control, and energy management [10, 11]. Despite this development, plant care in private households remains largely intuition-based.

This gap motivates the development of Plant Up!.

The primary objective of this thesis is to show that effective plant monitoring does not require expensive or specialized equipment. Using affordable, consumer-grade hardware such as soil moisture sensors, digital environmental sensors, and a microcontroller-based data acquisition system, this work demonstrates how common care mistakes can be reduced. In doing so, it aims to improve plant survival and provide users with a clearer understanding of the environmental conditions that influence plant health.

Based on the challenges outlined above, this thesis addresses the following research question:

How can a low-cost, sensor-based system be designed to provide objective environmental data and actionable care recommendations?

This question focuses on practical application rather than theoretical considerations. The goal is not to replace horticultural expertise, but to support everyday plant owners by providing objective information that simplifies decision-making. By translating sensor measurements into clear feedback, the system helps users recognize when action is necessary and when intervention would be counterproductive.

The relevance of this research can be viewed from several perspectives. From an environmental perspective, this project helps save water. Many people water their plants based on routine or guessing, which often leads to waste. By using sensors, water is only used when the plant actually needs it, supporting a more sustainable approach [12, 13].

From a practical perspective, many plant owners are frustrated when their plants die despite regular care. By visualizing environmental conditions and offering clear care recommendations, successful plant care becomes easier, as users do not need detailed botanical knowledge.

The topic is also relevant from a technological perspective. Advances in microcontrollers, sensor accuracy, and wireless communication make it possible to implement functional monitoring systems using affordable consumer-level hardware. This thesis demonstrates how such technologies can be combined into an accessible Internet of Things (IoT) solution for everyday plant care.

To understand why these measurements are essential, the following chapter examines the environmental factors that directly influence plant growth and health.

2.2 Environmental Factors in Plant Growth

Plants are living systems, but they can be treated as measurable input-output systems in engineering practice. Light, water, temperature, and humidity are not “nice-to-have” background conditions. They are measurable variables with operating ranges and failure limits. When a variable stays outside its range, stress is not a surprise. It is the expected outcome. This chapter focuses on what can be measured and controlled, because Plant Up! must convert plant needs into sensor values and thresholds. [14]

2.2.1 Importance of Soil Moisture, Temperature, Light, and Humidity

Soil moisture

Soil is a porous material made of solids plus pore space filled with air and water. An ideal soil for plant growth is often described as 50% solids and 50% pore space, ideally split into equal parts air and water so roots get both oxygen and moisture. This is the physical reason why “slightly moist” often performs better than “always wet.” [14]

When pore spaces stay waterlogged, air is displaced and the oxygen supply to roots drops. Root function declines and nutrient uptake suffers, even while the pot appears wet. This oxygen deficit is the primary mechanism behind overwatering damage and explains why soggy soil is more dangerous than dry soil for most indoor plants. [15]

The opposite end of the spectrum is the *permanent wilting point*: the threshold where remaining soil moisture is physically unavailable to roots. Below this point the plant cannot recover through normal watering, because the accessible water reserve is fully depleted. For monitoring, volumetric water content (VWC) expresses soil water as a percentage of total soil volume, which provides a consistent, loggable metric. [16, 17]

Light

Light is the energy input that drives photosynthesis. Plants do not use all wavelengths equally, so the relevant band is Photosynthetically Active Radiation (PAR), typically defined as the 400–700 nm range. This distinction matters because “bright to humans” and “usable for plants” are not the same thing. [18]

The plant-focused intensity metric is PPFD (Photosynthetic Photon Flux Density), which counts photons in the PAR range arriving per area per second. Many low-cost systems instead measure illuminance in lux, which is weighted for human vision and therefore does not map cleanly to PPFD. Lux is still practical for relative comparisons within one setup, such as comparing a dark corner to a bright window. [19]

Temperature and humidity

Temperature controls how fast plant processes run. When it is too low, growth slows as biochemical reactions become sluggish. When it is too high, regulation mechanisms lose efficiency and heat damage risk increases. For indoor monitoring, temperature is a core variable because it directly affects water demand and interacts with humidity. [14]

Humidity links to transpiration, the process where water evaporates from leaf surfaces and creates an upward flow that transports dissolved minerals. When humidity is low, transpiration accelerates and the plant loses water faster, potentially outpacing root uptake even if the soil is moist. Because temperature changes the air's water-holding capacity, humidity must always be interpreted together with temperature to correctly assess the atmospheric demand on the plant. [20, 21]

Technical link: translating needs into metrics

The described processes depend on physical quantities, not on perception. Roots react to oxygen availability and water content, not to how soil feels. Leaves respond to photon availability and atmospheric drying power, not to how bright or warm a room seems.

For control, these processes must be expressed as measurable variables. Soil moisture becomes VWC, light becomes PPFD or lux, and temperature and humidity define the atmospheric load on the plant. Only numerical values allow operating ranges and thresholds to be defined and monitored. However, simply tracking numbers is not enough. To design an effective warning system, one must understand what happens when these boundaries are breached.

2.2.2 Effects of Environmental Factors on Plant Growth

Deficiency: when values are too low

Low soil moisture reduces water availability until the permanent wilting point is reached. Monitoring must warn before the plant approaches this boundary, because recovery becomes unreliable once accessible water is depleted. [16]

Low light creates a different stress pattern. The plant tends to grow taller and thinner as it stretches toward available light, often producing weaker stems and paler leaves. This gradual change indicates chronic under-lighting and is not resolved by a single bright day. [22]

Low temperature slows growth by reducing reaction rates. Low humidity increases transpiration demand, which means the plant can become water-stressed even when the soil appears adequately moist. [21]

Excess: when values are too high

Excess soil moisture displaces air in pores, starving roots of oxygen. The plant can then show wilting symptoms even in wet soil, because damaged roots cannot absorb water properly. This is why overwatering often looks like underwatering to the untrained eye. Prolonged wet conditions also favor root decay pathogens, accelerating damage further. For engineering control, the upper moisture limit is a safety boundary that must be enforced with alerts. [15, 8]

Excess light and heat can cause leaf scorch, especially when a shade-adapted plant is suddenly exposed to strong sun. The damage occurs during short peak exposures rather than from long-term averages, which means monitoring must capture peaks, not only means. [23]

Safe corridors and thresholds

A monitoring system needs a control model that prevents both deficiency and excess. The simplest model is a safe corridor: a defined acceptable range for each variable. Inside the corridor, stress risk is low. Near corridor edges, the system should warn before the plant enters a damage trajectory. These corridors are species-dependent, so they must be configurable rather than hard-coded. [24]

A practical example: many houseplants prefer relative humidity around 40–60%, while tropical species often prefer higher levels. This is exactly why a system should start with reasonable defaults and then refine them using observed trends. Data makes the corridor adaptive instead of theoretical.

2.2.3 Why Sensor-Based Logging Matters

Human perception is qualitative, biased, and slow compared to plant dynamics. Surface touch checks miss root-zone conditions, the eye adapts to darkness and overestimates available light, and weekly observation misses hourly stress peaks. [25, 26] Continuous logging at fixed intervals is not a luxury but a practical requirement, because it reveals patterns and variability that single observations cannot capture. [21]

Sensors convert hidden processes into objective numbers and timestamps. They enable threshold checks, trend detection, and early warnings before visible symptoms appear. This is the real value of Plant Up!: plant care becomes a monitored system, not a guessing game.

With the biological variables defined and the necessity of objective measurement established, the next chapter evaluates the physical hardware required and describes how sensor signals are captured, filtered, and transmitted as structured data.

2.3 Sensor Technologies and the Measurement Pipeline

Every environmental variable, whether soil moisture, temperature, humidity, or light, must be converted into an electrical signal before a controller can process it. The choice of sensor determines not only what can be measured, but also how accurately and how long the system can operate without maintenance. But selecting appropriate hardware is only the first step. The raw signals must also be read, filtered, and packaged into structured data before they can be evaluated. This chapter covers both: the sensor selection for Plant Up! and the firmware-level data pipeline that turns their output into usable information.

2.3.1 Overview of Common Sensor Types

Soil moisture sensors

Low-cost soil moisture measurement is typically implemented using resistive or capacitive sensors.

A resistive sensor works by passing a small electric current between two metal probes inserted into the soil. Wet soil conducts electricity better than dry soil, so the measured resistance drops as moisture increases. The principle is straightforward and the hardware is cheap, but there is a significant disadvantage: the metal probes are directly exposed to wet soil, which causes them to corrode over time. As the probes degrade, readings become increasingly unreliable. [27]

A capacitive sensor takes a different approach. Instead of measuring how well the soil conducts current, it detects how the surrounding material affects an electric field generated by the sensor. In simple terms, water changes the electrical properties of the soil around the sensor, and this change is what gets measured. The key advantage is structural: the sensing element is embedded inside a coated circuit board, so no metal is directly exposed to the soil. This eliminates the corrosion problem and makes the sensor significantly more durable for long-term use. [28]

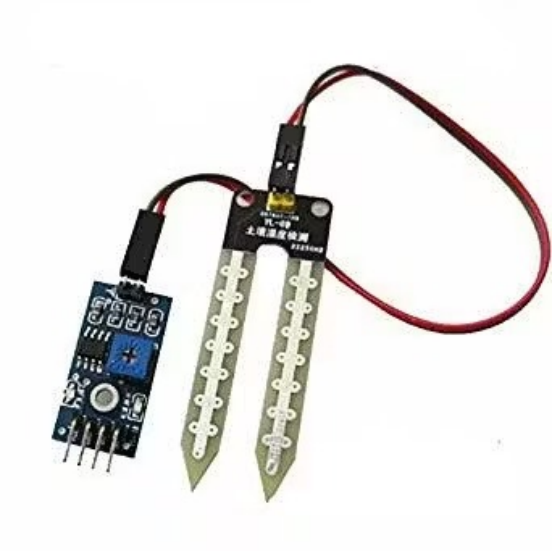

Figure 2.1: Resistive Soil Moisture Sensor [29]

Figure 2.2: Capacitive Soil Moisture Sensor [29]

As shown in Figures 2.1 and 2.2, the structural difference is immediately visible. The resistive sensor exposes two metallic forks directly to the soil, while the capacitive sensor uses a sealed PCB surface. This is the core reason why capacitive sensors are generally more suitable for long-term indoor monitoring. [27, 28]

Temperature and humidity sensors

Indoor monitoring often uses combined digital sensors such as the BME280. A digital sensor converts the measured value internally and sends a numerical result to the microcontroller, which eliminates the need to interpret raw analog voltages.

The BME280 communicates via I^2C, a two-wire protocol that uses one data line and one clock line. Multiple sensors can share the same two wires, which simplifies wiring considerably. For indoor plant monitoring, where detecting trends and crossing thresholds matters more than achieving laboratory-grade accuracy, the BME280 provides sufficient precision. [30, 31]

Light sensors

Light intensity can be measured using a photoresistor or a digital lux sensor. A photoresistor changes its resistance depending on brightness. It is simple and inexpensive but mainly useful for relative comparisons within the same setup. A digital lux sensor, by contrast, outputs illuminance directly in calibrated lux values. [32]

As discussed in the previous chapter, plants respond to light in the PAR range (400–700 nm), and lux is weighted for human vision rather than plant needs. [18] For basic indoor monitoring, however, lux values are still practical enough to detect whether a plant is receiving adequate or insufficient light.

All three sensor categories, moisture, climate, and light, offer viable solutions for low-cost monitoring. However, selecting the right sensor type is only half the equation. Understanding their practical limitations is equally important, because no sensor delivers perfect data under real-world conditions.

2.3.2 Limitations and Typical Error Sources

Non-uniform soil moisture distribution

Soil inside a pot does not dry evenly. Water moves downward due to gravity, and localized wet zones can persist long after watering.

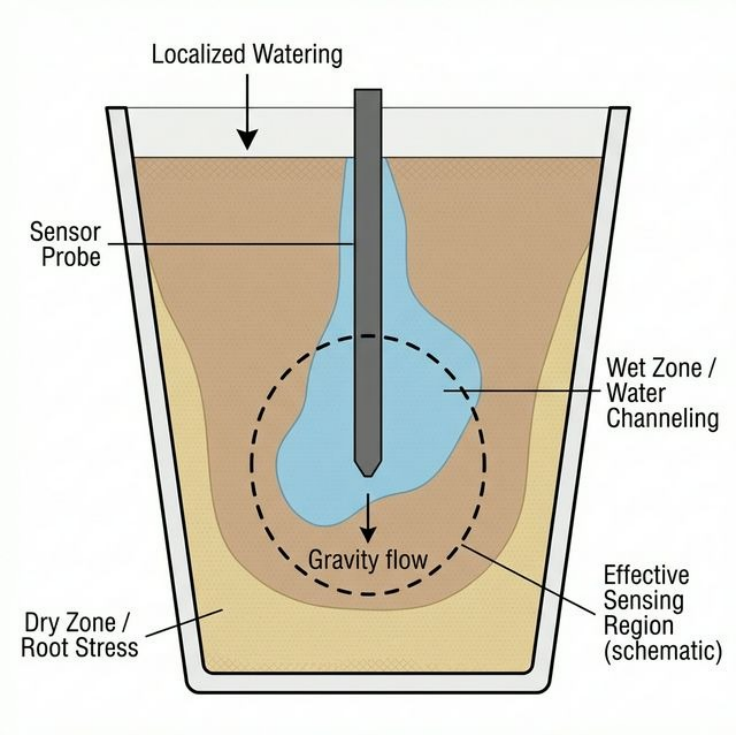

Figure 2.3: Measurement distortion caused by non-uniform soil moisture in a plant pot. The image illustrates how localized watering can distort sensor readings (AI-generated).

As shown in Figure 2.3, a soil moisture probe only measures a limited region around its body. A single reading may therefore not represent the entire pot. If the sensor happens to sit in a locally wet or dry pocket, threshold-based decisions can become misleading. Consistent sensor placement and insertion depth are therefore essential for reliable monitoring.

Sensor drift

Sensor drift describes a gradual shift in readings over time without any real change in the environment. For resistive sensors, corrosion is the primary driver: as the metal probes

degrade, the baseline resistance changes. Capacitive sensors reduce this problem significantly but can still drift due to aging or surface contamination. Periodic validation against a known reference improves long-term reliability. [27, 28]

Electrical noise and ADC limitations

Electrical noise refers to unwanted fluctuations in a signal caused by interference or unstable power supply. Analog sensors require an ADC (Analog-to-Digital Converter), which translates a continuous voltage into a discrete digital value. Because the resolution of the ADC is finite, very small moisture changes may not register at all. Hardware quality and basic software filtering, such as averaging multiple readings, therefore directly influence measurement stability.

2.3.3 Selection Criteria

For Plant Up!, a capacitive soil moisture sensor is the preferred choice. It avoids exposed metal electrodes, which eliminates the corrosion failure mode that limits resistive alternatives. This directly improves long-term stability, which is critical for a system intended to run continuously in a home environment. [28]

Compatibility with the ESP32 microcontroller is another practical constraint. The ESP32 operates at 3.3 V logic levels, so all selected sensors must support this voltage natively to avoid the need for additional level-shifting circuitry.

In this project, cost and durability outweigh the need for absolute calibration accuracy. The goal is reliable threshold monitoring, not scientific soil analysis. Knowing whether moisture is “inside the safe corridor” or “approaching the danger zone” matters more than knowing the exact volumetric water content to two decimal places.

Digital sensors such as the BME280 further simplify the hardware design. By communicating over I^2C, they reduce analog noise issues and allow multiple sensors to share a single data bus. [30]

Based on these criteria, the implemented hardware setup of Plant Up! consists of an ESP32 microcontroller, a capacitive soil moisture probe, and a BH1750 digital light sensor. The ESP32 serves as the central processing unit and provides integrated Wi-Fi connectivity as well as 12-bit ADC channels for analog measurements. Its 3.3 V logic level defines the voltage compatibility requirements for all connected sensors.

The BH1750 light sensor communicates via I^2C and directly outputs digital illuminance values in lux within a measurement range of approximately 1 to 65,000 lux. This avoids additional analog signal conditioning and simplifies integration.

The selected soil moisture probe is designed for 3.3 V operation and provides an analog voltage proportional to relative substrate moisture. Together, these components form the minimal stable sensing architecture of the system. A detailed component list with

supplier information is provided separately.

2.3.4 The Measurement Cycle

With the hardware selected, the remaining question is purely procedural: how does the controller read, clean, and transmit sensor values in a repeatable cycle?

Firmware read cycle

The firmware runs a repeating loop: wake, read sensors, process values, transmit, and sleep. Deep sleep, a power-saving mode where the CPU and most peripherals are shut down and only a small timer domain stays active for wake-up, is used after each transmission. [33]

The read sequence matters. The soil moisture sensor is analog, meaning its raw voltage needs a brief moment to stabilize after the controller wakes up. That is why it is read first. The BME280 and BH1750, on the other hand, are digital and deliver ready-to-use values over I^2C as soon as they are asked. Reading analog first and digital second keeps the cycle fast and stable. The entire active phase stays below approximately 2 seconds, after which the controller goes back to sleep and waits for the timer to start the next round. [33]

Sampling strategy and power management

In Plant Up!, the default measurement interval is 10 minutes, though it can be configured. This is long enough to keep battery consumption low but short enough to capture typical indoor changes such as watering events, sunlight shifts, or heating cycles.

The design relies on the extreme difference between active and sleep current. During active operation with Wi-Fi enabled, the ESP32 draws around 240 mA and can briefly peak above 300 mA during wireless calibration. [34] In deep sleep, current consumption drops to approximately 10 μA. [34] With timer-based automatic wake-up, this duty-cycling enables an estimated battery life of several weeks on a standard Li-Ion cell, depending on board design and signal quality. Shorter intervals, such as every 1–5 minutes, are supported for debugging and calibration but reduce runtime proportionally, because the Wi-Fi startup and transmission overhead becomes the dominant consumer. [33]

Signal processing and data cleaning

Each cycle takes multiple readings per sensor, typically five, and reduces them to one stable value. Simple averaging smooths random noise, while a median filter is useful when single spikes occur, for example from one bad ADC sample. This is especially

relevant for the analog soil moisture channel, where ADC fluctuations were already discussed in Section 2.3.2.

The firmware also performs plausibility checks. Physically impossible values are discarded, such as negative humidity or moisture readings above 100%. If a digital sensor does not respond within a timeout, the system logs a null value for that field and continues the cycle with the remaining sensors instead of blocking entirely. Null values are preserved in the payload so the backend can detect and handle data gaps.

Data packaging and transmission

After cleaning, values are assembled into a single JSON payload. JSON (JavaScript Object Notation) is a lightweight text format for structured data, built from name-value pairs. [35] A typical payload contains the device ID, a timestamp, and the current readings: soil moisture (%), temperature (°C), humidity (%), and light intensity (lux).

The payload is transmitted via MQTT, a lightweight publish/subscribe messaging protocol designed for constrained IoT devices. [36] How MQTT compares to REST in terms of latency and reliability is covered separately by Richard Onuoha in his chapter on IoT data synchronization in microservice architectures.

To keep the device robust in real apartments, where Wi-Fi can drop out or routers restart, the firmware includes retry logic. If the connection to the broker fails, the data point is buffered locally and retransmitted on the next cycle. No long-term storage is kept on the device. The intention is short buffering for connectivity gaps, not acting as a local database.

With this, the measurement cycle ends with clean, timestamped, structured values arriving at the backend. This is exactly the input needed for the next chapter, where thresholds and interpretation logic turn the raw data stream into actionable care recommendations.

2.4 Data-Driven Decision Making in Plant Care

After the measurement cycle described in Section 2.3, Plant Up! receives structured sensor data at the backend: timestamped values for moisture, climate, and light. This chapter explains how those incoming values are interpreted and transformed into feedback that a user can act on without needing to read raw numbers.

2.4.1 Threshold Zones and Plant-Specific Ranges

Building on the safe corridors defined in Section 2.2, each incoming sensor value is compared against an ideal range stored as a plant-specific configuration. The system classifies every parameter into three condition zones: Green (optimal, no action), Yellow

(close to a boundary, attention recommended), and Red (outside the safe range, action required).

This applies to soil moisture, temperature, humidity, light, and, if available, the electrical conductivity (EC) of the nutrient solution. The traffic-light mapping is intentionally chosen because green, amber, and red are widely recognized as low, medium, and high status codes in everyday decision aids.

Plant-specific ranges are not a universal default. A cactus is typically configured with lower moisture tolerance and higher light tolerance, while a fern prefers higher humidity and lower exposure to direct sunlight. In Plant Up!, these differences are handled as configurable parameters per plant profile rather than hardcoded constants.

Seasonal adjustment is implemented by storing separate summer and winter ranges. The reasoning is practical: indoor climate and light availability shift with seasons, and the same plant often needs different watering frequency and different tolerance bands depending on these conditions. Instead of forcing users to retune thresholds manually, Plant Up! switches between two predefined sets while keeping them editable.

To avoid overwhelming users with five separate decisions, Plant Up! computes a health score from 0–100% as a combined indicator. Such composite scoring simplifies interpretation without discarding the underlying data from individual sensors. [37] The health score compresses complexity into one overall status that is easy to scan. It is mapped to four labels: Great ($\geq$70%), Okay (40–70%), Needs Care (20–40%), and Critical ($<$20%).

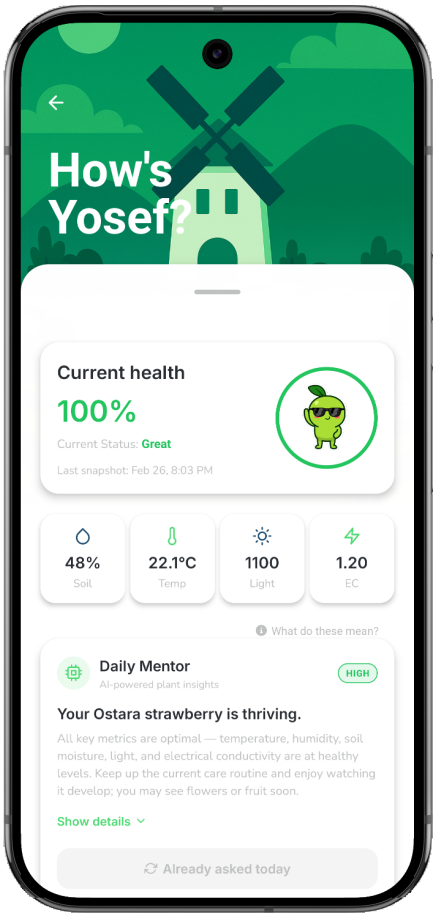

Figure 2.4: Plant status overview in the Plant Up! application showing health score, individual sensor values, and AI-generated care recommendations via the Daily Mentor feature.

As shown in Figure 2.4, the interface exposes both layers at once: the overall health score for quick orientation and the individual sensor values for users who want to verify the cause of a warning. The Daily Mentor feature is an AI-supported component that generates short natural-language hints based on the current zone states, turning combinations like “too dry” and “high temperature” into a concrete watering suggestion.

2.4.2 Deriving Care Recommendations

When a value leaves its ideal range, Plant Up! generates a recommendation phrased as an action, not as a measurement. Translating raw sensor readings into specific, user-facing actions is a well-established approach in threshold-based decision support systems. [38] The goal is that a user reads “Water it now” instead of interpreting “Soil moisture: 18%” under time pressure. The system evaluates parameters individually, and the first parameter that triggers a Yellow or Red state becomes the active recommendation until the next measurement cycle updates the situation.

Figure 2.5: Simplified decision flow: sensor data is compared against plant-specific thresholds and classified into condition zones, which then trigger the corresponding care recommendation (AI-generated).

Figure 2.5 summarizes this logic visually: incoming measurements are checked against stored thresholds, classified into zones, and then translated into one user-facing recommendation.

2.4.3 Visualization and Interpretation

Interpretation is not only logic. It is also presentation. Plant Up! uses a traffic-light visualization with color-coded indicators (green, yellow, red) and an animated mascot ("Sprig") that mirrors the current plant state. The purpose is speed: users should recognize "all good" versus "act now" in one glance, even without understanding the sensor units. Using color as a semantic status channel follows the principle that red signals danger, yellow signals caution, and green signals a safe condition, a convention standardized in ISO 3864. [39]

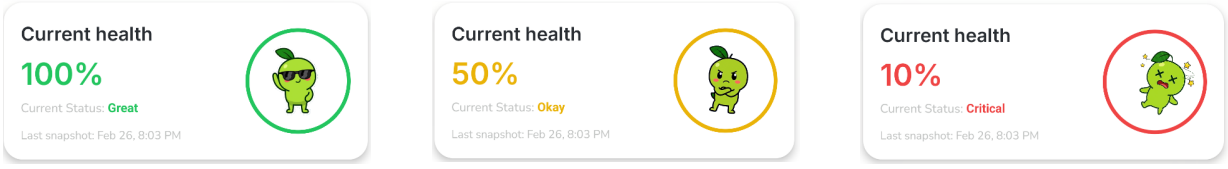

Figure 2.6: Traffic light health status in Plant Up!: optimal conditions (left, 100%), suboptimal conditions requiring attention (center, 50%), and critical state requiring immediate action (right, 10%).

Figure 2.6 illustrates these three primary health states. Rather than focusing on numbers, the interface leverages the mascot's expression and the background color to immediately communicate whether the plant is thriving, needs attention, or requires urgent care.

Single values are useful, but trends are where the "why" becomes visible. Plant Up! therefore shows historical data as bar charts with selectable time ranges (1 hour, 24 hours, 7 days, all time). Time-oriented visualizations make it easier to detect patterns and anomalies than isolated snapshots, because humans can quickly see slopes, cycles, and sudden jumps across a timeline. [40]

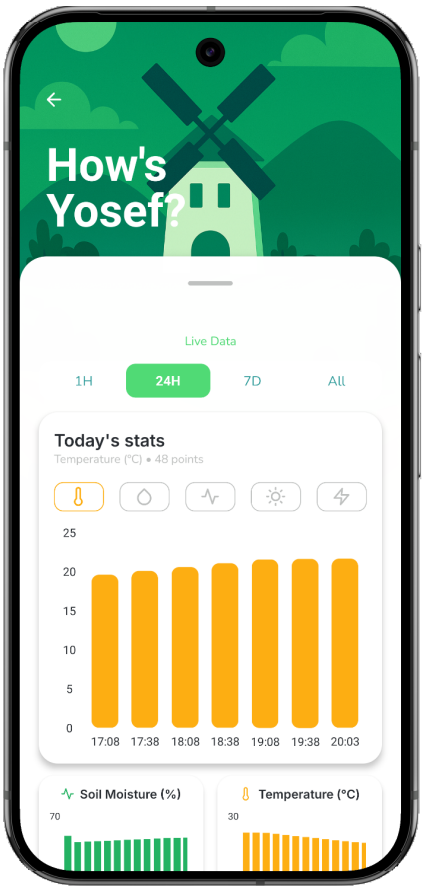

Figure 2.7: Historical sensor data visualization in Plant Up! showing temperature readings over a 24-hour period. Users can switch between different sensors and time ranges to identify trends and patterns.

As illustrated in Figure 2.7, a 24-hour chart can reveal repeating daily peaks, such as afternoon heating near a window, or unusual drops from night ventilation. These patterns support long-term care decisions like relocating the plant or adjusting the watering schedule proactively.

With these decision rules and visual mappings defined, Plant Up! can transform raw measurements into consistent feedback. The next chapter describes how this interpretation layer is implemented in the real system and tested under practical conditions.

2.5 System Implementation and Validation

This chapter describes the practical realization of the Plant Up! sensor station. With the sensor selection (Section 2.3.3), firmware pipeline (Section 2.3.4), and decision logic (Chapter 2.4) already established, the focus here is on three things that have not yet been covered: how the components were physically wired and housed, how raw ADC readings were calibrated into meaningful moisture values, and how the assembled system was validated against real plant data.

2.5.1 Hardware Assembly

- **Wiring and pin assignment:**
  - GPIO pin mapping table: which ESP32 pin connects to which sensor
  - Wiring diagram or schematic as figure
- **Enclosure:**
  - Custom 3D-printed housing for indoor use
  - Protection against water splashes, ventilation openings for climate sensor
  - Photo of the finished sensor station as figure

2.5.2 Sensor Calibration

- **Problem:**
  - ESP32 ADC produces raw values (0–4095) with no inherent unit
  - A user cannot interpret “ADC reads 2340” without context
- **Two-point calibration:**
  - Step 1: Record raw value in air-dry substrate (dry reference)
  - Step 2: Record raw value in fully saturated substrate (wet reference)
  - Linear mapping between these two endpoints to a 0–100% moisture scale
  - Concrete calibration values from the Plant Up! test setup
- **Limitations:**
  - Calibration is substrate-specific (different soil types produce different baselines)
  - Plant Up! provides relative moisture indication, not laboratory-grade VWC
  - Recalibration recommended when substrate or pot changes

2.5.3 Practical Validation

- **Test setup:**
  - Specific plant type, location, and monitoring duration
  - Controlled watering event during test phase
- **Expected vs. measured behavior:**
  - Soil moisture: spike after watering, gradual decline from evaporation and uptake (consistent with theory from Section 2.2)
  - Light: daily cycle visible (bright daytime, low at night)
  - Temperature and humidity: stable baseline with minor shifts from heating or ventilation
- **Evidence:**
  - At least one figure showing real sensor data from the Plant Up! app
  - Visual comparison between measured curve and expected physical behavior
- **Limitations:**
  - Limited number of test plants and short test duration
  - No laboratory reference sensor for absolute accuracy verification
  - Results validate functional correctness, not scientific precision

2.6 Conclusion

This thesis set out to answer a practical question: how can a low-cost, sensor-based system provide objective environmental data and actionable care recommendations for indoor plants? The chapters above have addressed this question step by step, from the biological foundations through sensor selection and firmware design to the interpretation logic that turns raw measurements into user-facing feedback.

The core argument of this work is that plant care does not need to rely on intuition. Soil moisture, temperature, humidity, and light are physical quantities with measurable ranges, and when those ranges are violated, the consequences are predictable. Chapter 2.2 established that both deficiency and excess cause specific, well-documented stress responses, and that human perception alone is too slow and too biased to detect these conditions reliably. Chapter 2.3 then showed that affordable consumer-grade hardware, specifically a capacitive soil moisture sensor, a BME280 climate sensor, and a BH1750 light sensor connected to an ESP32 microcontroller, can capture these variables with sufficient accuracy for threshold-based monitoring. The firmware pipeline described in Section 2.3.4 ensures that sensor data is read, filtered, and transmitted in a repeatable cycle with minimal power consumption.

Chapter 2.4 translated these raw measurements into decisions. By classifying each parameter into green, yellow, and red zones and combining them into a single health score, Plant Up! compresses complexity into feedback that requires no botanical knowledge to understand. The traffic-light system, the animated mascot, and the trend visualizations all serve the same purpose: making the plant's actual state visible at a glance.

The system is not perfect. The sensor station measures a single point in the pot, not the entire root zone. Calibration is substrate-dependent and produces relative values rather than absolute ones. The health score is a weighted approximation, not a scientific diagnosis. These are deliberate trade-offs. The goal was never laboratory-grade precision but a practical tool that prevents the most common care mistakes, especially overwatering, which remains the leading cause of houseplant failure.

By relying entirely on widely available consumer-grade components and open protocols, the system confirms the premise from the introduction: effective plant monitoring does not require specialized equipment or expert knowledge. All data remains under user control, and the architecture supports scaling to multiple plants through individual sensor stations reporting to a shared backend.

With the sensor station producing clean, timestamped measurements and the interpretation logic translating them into actionable feedback, the next question becomes: where should this data be processed, and how does it reach the backend? The following chapter by Abd-Ulrahman Behari addresses exactly this, comparing edge and cloud processing strategies for environmental sensor data in the context of smart gardening.

Kapitel 3

Edge vs. Cloud Processing in Smart Gardening: A Comparative Study for Environmental Sensor Data

Author: Abd-Ulrahman Behari

3.1 Introduction

Plant care? That is just watering and waiting!

This is how many people view gardening. However, the digitalization of houseplant care represents a growing intersection between the Internet of Things (IoT) and our daily lives. As urbanization increases and people spend more time indoors, the desire to maintain healthy plants has led to an emerging market of smart gardening devices. But many current solutions are isolated systems that only alert the user when a plant is already in critical condition. Does it really have to be this isolated?

The core idea for PlantUp originated during a team brainstorming session. We were discussing SStepUp,än application concept designed to let friends track each other's steps and provide mutual motivation. Simultaneously, we knew we wanted to build a physical system using ESP32 microcontrollers and environmental sensors. By combining the social, gamified motivation of tracking progress with IoT hardware, PlantUp was born. We aim to evolve the concept of gardening by integrating continuous sensor monitoring with a social platform, turning an isolated chore into a collaborative experience.

3.1.1 Microservices in Gardening

The future of gardening lies in digital transformation. It is no longer just about reading a moisture sensor; it is about connecting nature with technology. By sharing data across an enthusiast community, we enable a paradigm shift from individual plant care to data-

driven, collaborative gardening. But how can we build a backend infrastructure capable of handling continuous sensor streams and social interactions simultaneously?

[INSERT FIGURE: Concept Art of Social Gardening with IoT]

3.1.1.1 The Core Challenge: Latency and System Dependability

Sending sensor data across the continent just to update a dashboard? In a previous IoT project I worked on, environmental data was routed to a server in the Netherlands before it could be displayed on a website. The resulting delay was incredibly frustrating and highlighted a massive flaw in centralized systems. This personal experience proved that managing network latency (Round-Trip Time) and ensuring system dependability are not just theoretical concepts, but practical necessities for a smooth user experience.

These challenges form the architectural core of this thesis: navigating the trade-offs between Edge-Centric and Cloud-Centric paradigms to prevent exactly these types of delays. A detailed theoretical breakdown of these models is provided in Section 3.2.4.

3.1.2 Research Question

3.1.2.1 Primary Question

"What are the fundamental differences between Edge and Cloud processing within a microservices architecture, and what trade-offs do they present when implementing real-time monitoring, as demonstrated by the 'Plant Up!' ecosystem?"

In order to answer this, PlantUp serves as a comparative case study. This thesis will analyze the direct impact of logic placement on the final product, specifically comparing latency versus throughput, infrastructure cost versus complexity, data integrity, and overall system reliability.

3.1.3 Scope and Limitations

3.1.3.1 Computational Constraints of the Edge Device (ESP32)

Can a small microcontroller really make autonomous decisions? The ESP32 is capable, but it has strict processing limitations and a tight memory footprint. These constraints act as the primary architectural driver, forcing a careful balance between Edge-side data aggregation and Cloud-side decision-making.

Note: The specific network constraints, awake-time mechanics, and energy consumption profile of the ESP32 hardware are extensively detailed in Onuha's section (Chapter X). This section focuses strictly on how those physical limitations dictate the distribution of processing logic.

3.1.3.2 Comparison limited to Processing Location (Edge vs. Cloud), not Transport Protocols

To ensure a precise comparison, this thesis establishes clear scope boundaries. The analysis is strictly limited to the processing location (Edge computing versus Cloud computing). The transport layer is abstracted, with the MQTT protocol assumed as the baseline standard. Alternative protocols like CoAP or HTTP are excluded to keep communication variables fixed and the architectural comparison sharp.

3.2 Theoretical Framework

This section establishes the conceptual foundations necessary to understand scalable smart gardening systems. While building an IoT infrastructure presents significant complexity, standardized reference models provide a framework for structuring such systems. Therefore, this section outlines the required IoT architectures and explores modern cloud backend solutions like Backend-as-a-Service (BaaS). These theoretical building blocks provide the context for the concrete PlantUp architecture discussed in Chapter 4.

3.2.1 The IoT World Forum Reference Model (7-Layer Model)

To reason systematically about IoT architectures, this thesis utilizes the IoT World Forum 7-Layer Reference Model as a conceptual map. A layered architecture breaks a complex system into ordered levels, where each layer has a defined role and clear interfaces to the layers above and below it. While alternative models such as the OSI layer model exist, they are primarily oriented towards network communication protocols rather than the holistic integration of hardware, edge computing, and cloud infrastructure required for this use case. The 7-layer model serves as the optimal baseline for understanding the fundamental division between "Fog"(Edge) and Cloud computing.

In the IoT World Forum model, Layer 1 (Physical Devices & Controllers) represents the “edge” hardware such as ESP32 boards and their moisture, light, and temperature sensors, which directly interact with the physical environment. Layer 2 (Connectivity) covers the communication technologies like Wi-Fi, MQTT, and HTTPS that transport data between devices and backend systems.

Layer 3 (Edge or Fog Computing) contains processing that happens close to the data source, such as filtering, threshold checks, or basic aggregation on a local gateway or directly on the microcontroller. Edge computing is the practice of performing computation near where data is generated, instead of sending everything to a distant cloud, primarily to reduce latency and bandwidth usage. Layer 4 (Data Accumulation) focuses on storing data in transit and buffering it for further processing, for example via message brokers or initial ingestion into a managed database or BaaS platform. Data accumulation in this sense means persisting raw streams in a form that later layers can query reliably.

Layer 5 (Data Abstraction) transforms and structures stored data into more efficient and meaningful representations, such as aggregated views, hypertables in a time-series database, or standardized APIs serving cleaned data. Data abstraction therefore hides storage details and presents a consistent interface to application logic. Layer 6 (Application) contains the business logic and user interfaces, such as a React Native app, notification engine, or gamification components that interpret sensor data and present it to the user. Finally, Layer 7 (Collaboration & Processes) models how data drives human action and social interaction—for instance, sharing plant status with friends, participating in community leaderboards, or integrating plant health into broader workflows.

[Figure 3.1: The 7-Layer IoT World Forum Reference Model]

In this thesis, the central architectural question is how much decision logic should remain at Layer 3 (Edge) versus being shifted upward into Layer 5 (Data Abstraction) and Layer 6 (Application) in the cloud. The 7-layer model structures this discussion by establishing clear boundaries between physical sensing, connectivity, edge processing, and backend-centric gamification features.

3.2.2 Modern IoT Backends

The architecture of a modern IoT backend requires a robust foundation that can concurrently handle traditional relational data (like users and profiles) and the unique demands of high-velocity sensor streams. This section explores Backend-as-a-Service (BaaS) and specialized database extensions that address these requirements without the overhead of manually managing complex server infrastructures.

3.2.2.1 Backend-as-a-Service (BaaS): The Role of Supabase in IoT

A Backend-as-a-Service (BaaS) is a cloud platform that provides ready-made backend building blocks such as databases, authentication, file storage, and APIs, allowing developers to focus on application logic rather than server maintenance. For the PlantUp infrastructure, a dual requirement existed: the system needed robust relational data handling coupled with specialized capabilities for sensor streams and rapid client communication. Supabase was selected because it builds its architecture entirely around PostgreSQL, functioning as a Postgres-native backend. This approach combines the familiarity of SQL with managed infrastructure and critical functional extensions [168, 169, 170].

Specifically, Supabase satisfies two core operational requirements for the PlantUp ecosystem that traditional relational setups struggle with. First, it supports real-time subscriptions by exposing database changes over WebSockets. This real-time capability is essential for implementing the interactive, gamified elements of the platform, such as the social chatting features, by allowing clients and servers to push messages without opening a new HTTP request for each update. When a row in a subscribed table is inserted, Supabase instantly streams a change event to connected clients [169, 170, 168].

[Figure 3.2: Supabase Realtime Architecture Flow]

Second, the platform provides native integration with time-series database extensions. Relying purely on a standard relational system for continuous sensor data would require extensive, manual normalization efforts to accommodate varying sensor types, resulting in significant time expenditure and degraded query performance over time. By leveraging Supabase's integrated tooling, the BaaS model circumvents the necessity of manual normalization for high-velocity IoT data, making it feasible to iterate quickly on the social backend features while still maintaining a robust, scalable foundation [170, 168, 169].

3.2.2.2 Time-Series Databases: Why Standard SQL Fails for Sensor Streams

Relational databases were originally designed for transactional workloads, such as order processing or user management, where each query touches a manageable number of rows and data volumes grow at a moderate pace. For continuous sensor data, this assumption breaks down. High-velocity ingestion describes situations where a system must handle large amounts of incoming data per second—for example, frequent measurements from many sensors—and each new record must be validated, indexed, and written to disk. Under such loads, a traditional relational engine can become the bottleneck.[171, 172]

Most relational systems use a B-tree index, a tree-like data structure that helps the database find rows quickly by sorted keys, but which becomes increasingly expensive to maintain as tables grow very large. In a continuous stream, data never stops arriving; instead of occasional inserts, the system must keep accepting new rows while maintaining indexes and query performance. Over time, query degradation occurs: time-based queries that scan large ranges become slower as the table and its indexes expand, even if only recent data is typically needed. Storage bloat and data retention challenges appear when all historical sensor data is kept indefinitely in the same tables, making backups, index maintenance, and schema changes more and more costly. These scalability problems justify using specialized time-series extensions that optimize for time-based partitioning, compression, and efficient range queries rather than treating sensor data like ordinary transactional records.[172, 171]

3.2.2.3 The TimescaleDB Extension: Hypertables and Chunking Explained

TimescaleDB is a PostgreSQL extension that turns Postgres into a time-series–optimized database while keeping the SQL interface and tooling intact. An extension in this sense is an add-on module that plugs into PostgreSQL and extends it with new capabilities without replacing the core engine. TimescaleDB introduces the concept of a hypertable, which is a logical table that automatically partitions time-series data into smaller physical pieces while still appearing as a single table to the application. This automatic partitioning is implemented through chunking, where each chunk stores data for a specific time interval (and optionally an additional dimension like device ID).[173, 174, 175, 176]

Chunking exploits temporal locality, the observation that most queries and inserts target recent time intervals and therefore only touch a small subset of all stored data. By

mapping new inserts into the latest chunk and allowing queries to skip entire chunks outside the requested time range, TimescaleDB enables high-performance inserts and fast time-range queries even when total row counts reach billions. Continuous aggregates are precomputed, automatically refreshed summaries of time-series data (for example, hourly averages or daily minima) that trade some storage and compute for much faster analytical queries over long periods. A retention policy defines rules to automatically drop or compress old chunks after a configured time, so that storage growth remains controlled and the most relevant data stays fast to access.[174, 175, 176, 173]

[Figure 3.3: Hypertable vs. Normal Table Storage]

Compared to a normal relational table, a hypertable's internal chunk layout reduces the I/O and indexing burden on the database and lets developers keep using standard SQL while benefiting from time-series–specific optimizations.[176, 173, 174]

3.2.2.4 Object Storage and Media Handling in Microservices

Not all data in a smart gardening system is structured sensor telemetry. The gamified and social aspects of PlantUp necessitate the handling of user profile pictures, plant images, and potentially short videos. These files are unstructured or semi-structured and are fundamentally incompatible with high-performance relational database storage. Object storage is a service that stores files (objects) in buckets and references them by keys, optimized for durability, scalability, and large binary content instead of row-based queries [177, 178, 179, 180].

Integrating media directly into the main PostgreSQL database would have introduced severe performance bottlenecks, slowing down the critical time-series queries required for real-time sensor monitoring. Therefore, the architecture leverages Supabase Storage, an S3-compatible service that separates the Blob (Binary Large Object) data from relational metadata. Furthermore, utilizing the integrated Content Delivery Network (CDN) provided by the BaaS ensures that media assets are served from edge locations geographically closer to the end-user. This strict separation of concerns—using PostgreSQL exclusively for structured telemetry and user management, while delegating media to a dedicated object storage CDN—maintains database performance while delivering a responsive user interface [178, 179, 180, 177].

3.2.3 Microservices in IoT

As IoT systems grow, separating the backend into smaller, functional components becomes essential for both stability and development velocity. Microservices represent a distributed architectural style in which system functionality is split into small, independently deployable services, each responsible for a focused capability. This contrasts with monolithic backends where all logic is bundled into a single application and must be deployed as one unit. In the context of smart gardening, this style allowed the development team to separate device communication (MQTT), data storage (Supabase), and user-facing features into distinct, cooperating components. This separation forces

clear interfaces and prevents hardware-specific protocols from tangling with backend business logic [153].

3.2.3.1 The Gateway Pattern: Bridging Edge and Core Services

Architectural patterns are reusable, named solutions to recurring design problems. Within a microservices architecture, the Gateway pattern (often discussed as the API Gateway pattern) describes a service that acts as a single entry point between external clients—in this case, distributed ESP32 microcontrollers—and the internal backend services [153, 155, 154].

For PlantUp, establishing a dedicated Gateway explicitly to bridge the device world (MQTT) and the data world (Supabase) was a critical structural decision driven by security and scalability requirements. Directly connecting field-deployed ESP32 boards to the Supabase REST APIs would necessitate hardcoding database API keys directly onto the microcontroller firmware. This represents a severe security vulnerability, as hardware can be physically compromised or firmware extracted.

Instead, a middleware gateway acts as a secure buffer. A typical message broker receives published messages from devices and distributes them to subscribed clients based on topics, decoupling the publishers (ESP32 sensors) from consumers (Supabase). The gateway microservice subscribes to these topics and performs protocol translation, converting lightweight MQTT messages into structured database operations [154].

Beyond security, this decoupling provides crucial structural flexibility. As the user base scales, the backend infrastructure can be modified without requiring firmware updates across all deployed devices. Furthermore, it enables regional scalability; additional MQTT brokers can be deployed in different geographic regions to minimize device-to-broker latency, while the gateway continues to securely funnel all translated telemetry into the central Supabase instances [156, 154].

In summary, the Gateway pattern provides a controlled entry point where all device traffic is validated, shaped, and translated before it reaches the core database. This avoids the security risks of hardcoded API keys while ensuring the architecture remains highly modular and geographically scalable [154].

3.2.3.2 Service Inter-communication: Synchronous vs. Asynchronous Patterns

Inter-service communication describes how microservices exchange data to fulfill user requests or process device telemetry. Synchronous communication requires the calling service to pause its execution (block) while waiting for the receiving service to reply, typically using HTTP/REST or gRPC. While this provides immediate consistency—ensuring that the caller knows exactly when an operation has succeeded—it introduces dangerous coupling. If a downstream service is slow, the upstream caller is delayed, creating a cascade of blocking operations [157, 158].

To mitigate these blocking bottlenecks, modern web servers and IoT gateways heavily rely on asynchronous communication. In an event-driven, asynchronous model, a service dispatches a message to a queue or broker and immediately resumes its own work without waiting for a direct reply. For an IoT platform handling multiple simultaneous users and continuous sensor data streams, this non-blocking behavior is paramount. Within the PlantUp architecture, the ingestion of sensor data via the MQTT gateway is inherently asynchronous; the ESP32 publishes a reading and is free to return to sleep, while the backend processes the telemetry at its own pace. This design ensures that traffic spikes from thousands of devices do not crash the entry nodes, trading immediate consistency for high availability and resilience [159, 158, 157, 160].

In summary, microservices allow IoT backends to be decomposed into focused, independently deployable services, while the Gateway pattern introduces a controlled entry point that isolates internal logic from direct device communication. At the same time, the choice between synchronous and asynchronous inter-service communication directly shapes latency, reliability, and coupling, which are exactly the dimensions that matter when deciding how much logic to keep at the edge versus offload to the cloud in a smart gardening system. These concepts form the microservice perspective that the next section will apply to the concrete PlantUp architecture.[157, 153, 158, 154]

3.2.4 Architectural Paradigms: Edge vs. Cloud

Architectural design in IoT systems is fundamentally defined by the locus of computation. Deciding whether logic resides in the cloud or directly on the microcontroller dictates latency, reliability, and power consumption constraints.

3.2.4.1 Defining the Cloud-Centric Model

A cloud-centric architecture concentrates the computational load, long-term data storage, and decision-making within remote data centers. The devices at the edge act as "dumb sensors,"reading raw environmental values and immediately forwarding them to the cloud for validation, aggregation, and interpretation [182, 170].

The primary advantage of this centralization is raw computational power and horizontal scalability. Complex queries, time-series aggregations, and social features can be executed without concern for the strict hardware limitations of an embedded device. However, a purely cloud-centric model is entirely dependent on continuous internet connectivity and is subject to network latency (Round-Trip Time). While minor delays might be acceptable for periodic temperature logging, they can disrupt real-time control loops or immediate user feedback mechanisms [183, 184, 182, 170].

3.2.4.2 Defining the Edge-Centric Model

Conversely, an edge-centric architecture pushes computation, data aggregation, and autonomous decision-making down to the local device. The microcontroller is responsible for interpreting its own readings, evaluating thresholds, and triggering local actuation (such as controlling an LED or a water pump) without waiting for cloud instruction [185, 186, 170].

This model guarantees ultra-low latency and offline fault tolerance. If the external network fails, the device can still protect the plant. Furthermore, bandwidth usage is minimized since only summarized events, rather than raw streams, are transmitted outward. However, moving complex logic to the edge introduces severe development challenges. Firmware updates become complex deployment operations, remote debugging is highly restricted, and the code must operate within the strict CPU, RAM, and battery constraints of microcontrollers like the ESP32 [182, 185, 170, 186].

[INSERT FIGURE: Visual Comparison of Cloud vs. Edge Architectures]

3.2.4.3 The Spectrum of Logic Distribution: “Thick Edge” vs. “Thin Edge”

In architectural practice, the division between Edge and Cloud exists on a continuous spectrum. A “Thick Edge” delegates substantial processing and autonomous control to the microcontroller, prioritizing offline robustness over device simplicity. A “Thin Edge” relies heavily on the cloud, utilizing the microcontroller strictly for sensing and transmitting raw or minimally processed data.

To optimize the PlantUp ecosystem, an explicit decision was made to implement a Thin Edge architecture. The ESP32 units within the system are constrained strictly to low-level hardware interactions: calibrating the sensors, executing the analog-to-digital conversions, and structuring the resulting data into a JSON payload for transmission. All complex logic, database aggregation, threshold checking, and social gamification are offloaded entirely to the Supabase backend and the React Native frontend application.

This deliberate placement on the logic distribution spectrum minimizes firmware complexity and maximizes the flexibility of the cloud services. A detailed technical breakdown of how this Thin Edge paradigm is implemented in practice—tracing the data from the ESP32 sensors through the MQTT gateway and into the TimescaleDB storage—forms the focus of Chapter 4: Architecture and Implementation [170, 181].

3.3 The PlantUp Architecture

PlantUp translates the theoretical paradigms established in Section 3.2 into a concrete, deployable architecture for a socially integrated smart-gardening ecosystem. The system design strictly adheres to the IoT World Forum layered model, decomposes monolithic backend logic into focused microservices, and utilizes a robust gateway pattern to

securely manage device communication. Crucially, the architecture deliberately implements a “Thin Edge” paradigm (as concluded in Section 3.4.3), offloading complex logic to the cloud to prioritize computational flexibility and long-term analytics capabilities.

3.3.1 System Overview

The architecture connects a physical, sensor-equipped edge device with a scalable cloud backend, ultimately serving synthesized data to a React Native mobile frontend. This foundational pipeline was designed to ensure continuous remote accessibility and structural scalability.

3.3.1.1 High-Level Architecture Pipeline

At the physical layer, PlantUp utilizes an ESP32 microcontroller deployed directly within the plant environment. The ESP32 interfaces with attached analog and digital sensors (soil moisture, temperature, humidity, and light intensity), periodically structuring the raw environmental measurements into standardized payloads. In the terminology of the 7-Layer IoT model, these components constitute Layer 1 (Physical Devices) and Layer 3 (Edge Computing).

A critical design decision in this pipeline was the exclusive reliance on Wi-Fi and MQTT for device-to-cloud communication (Layer 2 - Connectivity), consciously avoiding direct device-to-smartphone technologies such as Bluetooth Low Energy (BLE). While BLE offers localized energy efficiency, it inherently restricts real-time accessibility to the immediate physical vicinity of the device and introduces persistent user-experience friction through frequent pairing bugs and required reconnections. The Wi-Fi-to-Cloud pipeline ensures a seamless “connect once” onboarding experience, allowing users to monitor plant health globally without deploying a dedicated master translation gateway in their home. A detailed quantitative performance evaluation validating the choice of MQTT over traditional synchronous REST for this pipeline is presented by Onuha in the section *Experimental Evaluation of MQTT vs. REST*.

The central infrastructural tier is the Supabase cloud backend, functioning as both Layer 4 (Data Accumulation) and Layer 5 (Data Abstraction). To proactively architect for future scalability, the database is intentionally partitioned into two strictly distinct logical schemas: a *Social* schema governing users, gamification mechanics, and interpersonal posts, and a *Micro* (telemetry) schema dedicated solely to device state and high-velocity time-series sensor data.

This dual-schema segregation directly supports the long-term migration toward a “Database-per-Service” microservice architecture (Section 3.3.1) by ensuring data domains do not become tightly coupled. Furthermore, it significantly reduces query complexity for the React Native frontend application (Layer 6 and 7); specific screens can independently retrieve necessary UI data from the Social schema without parsing heavy telemetry records. End-to-end, this continuous pipeline guarantees that sensor measurements generated by the ESP32 securely traverse the MQTT gateway, are intelligently partitioned

within the Postgres database, and are immediately available for the mobile application to visualize and act upon.

[INSERT FIGURE: High-Level System Architecture Diagram]

The high-level diagram visualizes this continuous ingress: Sensors → ESP32 (Wi-Fi) → Cloud Backend (Supabase) → React Native App, explicitly illustrating the separation between the telemetry (Micro) and application (Social) schemas.

3.3.2 The Microservices Layer (Cloud)

PlantUp's cloud backend operationalizes the microservice principles introduced in Section 3.2.3. Rather than deploying a monolithic architecture where a single application instance governs both hardware ingestion and social networking features, the backend is strictly partitioned into specialized logical services. Every service is built around a specific domain concern—such as gamification mechanics, edge device management, or artificial intelligence processing—and communicates exclusively via well-defined interfaces. This structural separation is critical for system maintainability, enabling independent service evolution and robust fault isolation.

3.3.2.1 Application Layer Services: Social Networking and Gamification

Within the upper application layer, PlantUp deploys a suite of services dedicated exclusively to user experience and social integration. A centralized social feed service orchestrates user posts, interpersonal comments, and the distribution of shared plant statuses. Media assets, such as user-uploaded plant photography, are managed through Supabase Storage, linking object storage pointers with robust database access controls.

Parallel to the social feed, a distinct gamification engine continuously tracks and calculates user experience points (XP), growth milestones, and interaction streaks. Another highly specialized component is the "Daily Mentor", an AI-driven service that asynchronously processes recent sensor data to generate actionable gardening insights. Furthermore, a dedicated authentication and profile service comprehensively handles user identity and session management.

A foundational architectural decision in this layer was the strict adherence to the Single Responsibility Principle (SRP). Auxiliary features like the Daily Mentor and the gamification engine are explicitly decoupled from the core plant monitoring infrastructure. This ensures horizontal fault isolation: should the AI text-generation service experience high latency or catastrophic failure, it will strictly impact the Daily Mentor functionality, while users will seamlessly retain full, real-time access to their critical soil moisture and temperature telemetry. These services operate consistently across Layer 6 (Application) and Layer 7 (Collaboration & Processes) of the IoT model, highlighting the necessity of a cloud-centric architecture for a socially driven product where features inherently depend on shared, cross-user persistent state.

3.3.2.2 Device Management Service (Gateway-Oriented Isolation)

Operating perpendicularly to the user-facing application services is the centralized Device Management Service. This component serves as the concrete, implementation-level realization of the Gateway pattern discussed in Section 3.2.3, functioning as an absolute abstraction barrier between the volatile hardware domain and the structured application domain.

The primary architectural mandate of this service is to enforce a strict boundary between the time-series optimized telemetry data (TimescaleDB) and the standard relational social data (PostgreSQL). It is structurally dangerous to permit social feeds or gamification engines to execute raw SQL queries directly against high-velocity, multi-million-row sensor tables. Instead, the Device Management Service ingests, centralizes, and normalizes the unrefined device state—tracking connectivity status, hardware registration, and sensor calibration metrics.

By absorbing the raw MQTT payloads, this gateway service translates device-originated time-series data into sanitized, consistent backend structures. It subsequently exposes a simplified, read-optimized API interface. The gamification and social services interact solely with this abstracted API to fetch current plant conditions, entirely shielding them from the underlying low-level protocol specifics, TimeScaleDB chunk management, and edge-node lifecycle anomalies. This rigorous gateway-oriented isolation ensures that edge connectivity failures or malformed sensor telemetry cannot cascade into the application layer, thus preserving backend stability.

3.3.3 The Data Persistence Layer (Supabase)

To securely manage relational application state, high-velocity time-series telemetry, and user-generated media objects, PlantUp utilizes a consolidated PostgreSQL-backed environment via Supabase. This infrastructure aligns with the theoretical constructs outlined in Section 3.2.2, positioning a Backend-as-a-Service (BaaS) platform to deliver both traditional relational modeling and advanced time-series extensions within a unified, managed instance.

3.3.3.1 Unified Infrastructure with Schema-Level Isolation

While maintaining a single infrastructural source of truth simplifies initial deployment and cross-schema query execution, PlantUp deliberately isolates its data domains. The database is strictly partitioned into distinct logical schemas: a social schema encompassing users, plants, and interaction metrics, alongside a telemetry (micro) schema wholly dedicated to raw and derived environmental sensor logs.

This schema-level isolation was architecturally designed with explicit foresight for future horizontal scalability. By entirely compartmentalizing the TimescaleDB hypertables within the telemetry schema, the architecture seamlessly supports an eventual migration

toward a true “Database-per-Service” model. Should the sheer volume of MQTT device telemetry eventually exceed the performance capacity of the shared Postgres instance, the telemetry schema can be extracted and migrated to a standalone database cluster with zero structural changes required to the relational social schema. Currently, however, hosting both schemas within a unified Supabase instance permits centralized row-level security policies, consistent API abstraction, and efficient cross-schema queries—such as joining immediate soil moisture metrics with user-defined alert thresholds.

3.3.3.2 Time-Series Performance Optimization via TimescaleDB

A critical requirement of the PlantUp React Native application is the near-instantaneous retrieval of the most recent environmental data for active plant monitoring. Standard relational tables, while effective for discrete entity relationships, suffer from severe index bloat and degraded read performance when subjected to the high-frequency insert and retrieval rates typical of continuous sensor telemetry (as discussed in Section 3.2.2.2).

To address this, sensor measurements—including soil moisture, ambient temperature, and light intensity—are ingested directly into TimescaleDB hypertables, an extension operating natively within the Postgres environment. TimescaleDB structurally partitions this continuous stream of data into contiguous temporal chunks. This architecture effectively bypasses the performance bottlenecks of deep B-tree indices on massive tables by maintaining shallow indices within smaller time windows.

Because the mobile application frequently requests data specifically from the most recent chronological window (e.g., “the last 24 hours of moisture trends”), these queries are executed almost entirely against in-memory active chunks, resulting in drastically reduced query latency. This structural optimization is absolutely essential for the fluidity of historical charts and the rapid generation of Daily Mentor insights. By ensuring that analytical time-series workloads do not lock or degrade the performance of operational workloads (such as social feed interactions), PlantUp’s persistence layer successfully implements advanced time-series theory while maintaining the structural flexibility required for long-term project evolution.

3.4 Methodology

This section describes the experimental design used to evaluate two architectural approaches for PlantUp: a Cloud-Centric variant with a Thin Edge, and an Edge-Centric variant with a Thick Edge. Both variants run on the same ESP32 hardware and use the same Supabase-based backend stack so that the only relevant difference is where the decision-making logic is executed. By keeping hardware, transport, and storage technologies constant, the experiments isolate the impact of logic distribution on latency, resilience, and database behavior.

3.4.1 Experimental Architectures

Two architectural variants were implemented to directly compare how system behavior changes when core logic is placed in the cloud versus on the edge device. The variants follow the conceptual models defined in Section 3.2.4, with Architecture A representing a Cloud-Centric, Thin Edge approach and Architecture B representing an Edge-Centric, Thick Edge approach. The primary variable is the location of threshold evaluation and condition logic; everything else is deliberately kept as similar as possible.

3.4.1.1 Architecture A: Cloud-Centric (The "Connected" Plant)

Architecture A reflects the Cloud-Centric (Thin Edge) model from Section 3.2.4.1. In this setup, decision-making is centralized in the backend, while the ESP32 focuses on collecting and forwarding measurements. The edge device periodically reads environmental sensors (soil moisture, temperature, humidity, light) and forwards them to the cloud as JSON-structured payloads. Each payload includes a measurement type, a numeric value, and a timestamp, optionally enriched with a device identifier.

On the server side, the backend performs threshold evaluation and condition logic. Incoming measurements are written into TimescaleDB-backed hypertables for historical storage, and server-side rules determine whether the current state requires action (for example, "soil moisture too low"). Notification latency is measured by recording the time between a sensor reading crossing a threshold and the corresponding push notification being delivered to the React Native app. Because all logic is in the cloud, this architecture depends strongly on network bandwidth and connectivity but simplifies device firmware and centralizes control and analytics.

[INSERT DIAGRAM: Architecture A Data Flow (Sensor $\rightarrow$ Cloud $\rightarrow$ Push Notification)]

The diagram for Architecture A shows raw sensor data flowing from the ESP32 to the cloud backend, where logic is executed and push notifications are then delivered to the user.

[INSERT CODE SNIPPET: Cloud-Centric Logic Implementation]

This code placeholder marks where the concrete cloud-side logic (for example, threshold evaluation and notification triggers) will be documented later.

3.4.1.2 Architecture B: Edge-Centric (The "Smart" Plant)

Architecture B reflects the Edge-Centric (Thick Edge) model from Section 3.2.4.2. Here, intelligence is distributed, and the ESP32 performs local threshold comparison and autonomous decision-making. The device reads sensor values, calculates derived metrics such as moisture percentage, and compares them against locally stored thresholds. Immediate local status indication, such as an LED or other indicator, is triggered directly

on the device without waiting for a cloud response.

Instead of sending every raw measurement, the ESP32 logs the results of its decisions to the backend as JSON payloads. These logs might contain aggregated values, decision outcomes, or compact summaries rather than full raw streams, which introduces an eventual consistency model: the backend may temporarily lag behind the device state but catches up as logs arrive. Server-side processing is reduced because much of the evaluation has already happened on the device. This architecture is more tolerant to disconnects—local behavior continues even if the backend is unavailable—while still maintaining a historical record in the database once connectivity returns.

[INSERT DIAGRAM: Architecture B Data Flow (Sensor → Edge → Local Decision / Async Log)]

The diagram for Architecture B highlights that primary decisions are made on the edge path, with asynchronous logging to the cloud for persistence and analytics.

[INSERT CODE SNIPPET: Edge-Centric Logic Implementation]

This code placeholder marks where the device-side decision logic will be described in detail later.

3.4.2 Implementation Details

Both architectures share the same data structure and backend interface. The ESP32 in each variant sends JSON documents to the same Supabase endpoints, and the backend stores data in the same schemas and tables. The only intentional difference lies in where evaluation happens: Architecture A executes most logic in the cloud, while Architecture B executes it on the device and treats the backend primarily as a logger and analytics platform.

3.4.2.1 JSON Data Structure

Sensor readings are serialized as JSON objects to ensure a simple and language-agnostic interface. Each object contains at least the following fields: a measurement type (for example, “soil_moisture” or “temperature”), a numeric value, a timestamp representing when the measurement was taken, and a device identifier that links the reading to a specific controller. This uniform structure enables consistent backend processing, because database ingestion and validation logic can treat Architecture A and Architecture B payloads in the same way. The same schema is used in both variants, which keeps database design and querying identical across experiments.

3.4.2.2 Backend Integration

On the backend, both architectures integrate with Supabase using a shared access pattern. The system uses client libraries to write JSON data into Postgres tables, with realtime subscriptions for live updates and database triggers for secondary processing such as notifications or aggregation. State synchronization between device logs and user-facing views is handled through these triggers and subscriptions, allowing the React Native app to present near-real-time status changes. Historical querying is performed via TimescaleDB features on the same unified schema, ensuring that architectural differences do not require different query strategies.

3.4.3 Test Scenarios

To compare the two architectural paradigms fairly, controlled experiments were designed around crucial system metrics: latency, scalability, and database resilience. The foundational hypothesis underpinning this project favored the Cloud-Centric (Thin Edge) architecture due to its stark reduction in device-side firmware complexity and superior maintainability. However, it was necessary to empirically quantify the inevitable performance degradation introduced by cloud round-trip latency and determine if it remained within acceptable operational bounds. Given the domain context—botanical physiology—environmental metrics such as soil moisture do not necessitate sub-second, real-time reactivity. Intermittent state snapshots captured every 5 to 10 minutes provide entirely sufficient granularity for effective plant monitoring. Therefore, the goal of these test scenarios is strictly to ensure that the Thin Edge penalty does not violate these broader conceptual tolerances. The same measurement scripts and dashboards are used for both Architecture A and Architecture B to ensure rigorous comparability [170].

3.4.3.1 Scenario 1: Baseline Operational Conditions

Scenario 1 serves as the benchmark reference. The system is evaluated under stable Wi-Fi connectivity with standard network latency, absent of any artificial load injection. The objective is to record the native baseline measurements for notification latency, end-to-end response time, and database write throughput under optimal, "healthyöperational conditions. These baseline results establish the comparative foundation for the subsequent stress testing.

3.4.3.2 Scenario 2: Infrastructure Stress Test (1,000 Concurrent Writes)

Scenario 2 evaluates system scalability and database resilience under intense transactional stress. Unlike the steady, 5-to-10-minute snapshot intervals expected during normal operation, this scenario subjects the backend API to a sustained ingestion burst of exactly 1,000 concurrent write operations. This specific volume was deliberately selected based on project projections: the PlantUp ecosystem is architecturally designed to support an expected minimum baseline of 1,000 daily active users. Simulating 1,000

concurrent writes therefore represents a realistic, peak-traffic stress test for the provisioned infrastructure.

By simulating massive, synchronized burst traffic (resembling thousands of devices simultaneously reconnecting and finalizing queued payloads after a widespread network outage), the experiment observes resulting latency spikes, backend resource contention, and how swiftly the ingestion queues normalize once the peak subsides. The resulting data dictates how each architectural variant degrades under extreme pressure, and whether the unified database strategy requires structural scaling adjustments to accommodate real-world deployment growth.

[INSERT PICTURE: Load Testing Setup / Console Logs]

The accompanying figure illustrates the load-testing environment, specifically displaying representative console logs and the Supabase monitoring dashboards during the ingestion spike.

3.5 Results and Data Analysis

3.5.1 Latency Analysis

3.5.1.1 Edge Response Time (<50ms) vs. Cloud Round Trip (>200ms + DB Query Time)

- Quantitative analysis, edge speed, sub-50ms response, cloud delay, >200ms latency, network overhead, database query time, round-trip metrics, comparison charts, performance penalty, real-time constraints, user experience impact
- **[INSERT GRAPH: Comparison Bar Chart of Alert Generation Time (Edge vs. Cloud)]**
- Source: `https://ifactoryapp.com/blog/real-time-ai-cloud-latency-manufac`

3.5.2 Database Performance

3.5.2.1 Impact of High Ingestion Rates on C# Service Query Performance

- Performance impact, ingestion throughput, query latency, lock contention, C# service load, resource usage, CPU/Memory metrics, read-write conflict, optimization analysis, database tuning, concurrency issues
- **[INSERT GRAPH: Query Latency vs. Ingestion Rate]**

3.5.2.2 Storage Efficiency: Raw Data (Cloud Decision) vs. Batched Aggregates (Edge Decision)

- Storagemetrics, raw data volume, aggregation efficiency, payload size, storage costs, bandwidth consumption, cloud storage, batched transmission, data compression, economy of scale, archival strategy
- **[INSERT CHART: Projected Storage Costs Over Time]**
- Source: `https://metadeskglobal.com/from-raw-sensor-data-to-intelligent-a`
  `https://stonefly.com/blog/iot-data-storage-architectures-databases/`

3.5.3 Reliability and Resilience

3.5.3.1 Alert Delivery Success Rates during a simulated 1-hour internet outage

- Reliability metrics, delivery success analysis, outage simulation, local data buffering, resync capability, data integrity, critical failure, connectivity loss, edge resilience, robust design, message queuing
- **[INSERT TIMELINE: Data Sync Recovery after 1-Hour Outage]**
- Source: `https://www.opal-rt.com/blog/5-essential-computer-simulation-tec`

3.5.4 Gamification Synchronization

3.5.4.1 The "Lag" between Action (Watering) and Reward (XP Update in Social Service)

- User experience, feedback delay, action-reward loop, gamification lag, sync latency, psychological impact, interface responsiveness, XP updates, social satisfaction, system consistency, near real-time
- **[INSERT DIAGRAM: Sequence Diagram of Action-to-Reward Delay]**
- Source: `https://supabase.com/docs/guides/functions`

3.6 Discussion

The experimental evaluations conducted in the preceding chapter transition the theoretical trade-offs between Edge-centric and Cloud-centric architectures into concrete, empirical data. Rather than attempting to force a universal architectural conclusion, the findings contextualize these paradigms specifically within the domain of consumer

smart-gardening. The following subsections synthesize these results to justify the final architectural trajectory of the PlantUp ecosystem.

3.6.1 The "Thin Edge"Justification for PlantUp

3.6.1.1 Prioritizing Centralized Analytics over Edge Autonomy

The empirical measurements consistently demonstrate that for the specific functional requirements of PlantUp, a Cloud-centric (Thin Edge) architecture is not merely sufficient, but optimal. The fundamental domain constraint—botanical physiology—dictates that environmental metrics such as soil moisture and ambient temperature do not fluctuate rapidly enough to necessitate millisecond-level actuation loops. Consequently, the localized autonomy offered by a Thick Edge design provides negligible functional benefit to the end user.

Instead, deliberately stripping the ESP32 firmware of complex threshold evaluation logic vastly simplifies device provisioning, reduces the micro-controller's processing overhead, and minimizes the payload size transmitted via MQTT. All decision-making, including the generation of push notifications and the orchestration of the gamification engine, is centralized in the Supabase backend. This concentration of logic directly enables the rich, cross-user social features and the Daily Mentor AI integrations, which inherently require continuous access to historical datasets and aggregate user interactions. While this approach inevitably introduces network latency and mandates internet connectivity for logical evaluation, the experiments confirm that this latency remains entirely imperceptible within the context of hourly or sub-hourly plant care.

3.6.2 Database Integration and Client Architecture

3.6.2.1 Direct-to-Database Client Architecture via Supabase

From a frontend integration perspective, the experiments validate the architectural decision to utilize Supabase as a unified Backend-as-a-Service (BaaS). By leveraging the official Supabase JavaScript SDK, the React Native application establishes a direct, secure communication channel with the Postgres database.

This explicit reliance on the provider's SDK fundamentally eliminates the engineering overhead required to construct, deploy, and maintain a custom intermediary REST API layer (e.g., a dedicated Node.js or Spring Boot middleware). The client application authenticates directly through Supabase Auth, passes the securely scoped JWT (JSON Web Token), and subsequently executes queries against the database. The structural integrity and security of this direct-access model are enforced entirely through PostgreSQL's native Row Level Security (RLS) policies, ensuring that users can only query or mutate their specific social and telemetry data [194, 195, 196, 188].

3.6.2.2 Analytic Specialization vs. Real-Time Actuation

The architectural evaluation further confirms the explicit partitioning of database responsibilities. The use of TimescaleDB within the backend is highly optimized for the rapid ingestion of MQTT payloads and the execution of aggregated historical queries (e.g., rendering seven-day moisture trends in the mobile app). However, the literature correctly asserts that time-series databases introduce inherent query latency and connection overhead that makes them unsuitable for strict, real-time control loops [197, 198, 199, 189]. For PlantUp, this limitation is architecturally irrelevant. Because the system does not actuate physical hardware (such as triggering an automated water pump with sub-second precision), the asynchronous nature of cloud-based threshold evaluation against TimescaleDB is entirely acceptable and aligns perfectly with the Thin Edge strategy.

3.6.3 Complexity vs. Operational Cost

3.6.3.1 Minimizing Edge Complexity to Accelerate Development

A core theme emerging from the architectural comparison is the relationship between developmental complexity and operational cost. Attempting to maintain conditional logic simultaneously on the edge (C++ on the ESP32) and in the cloud (SQL/Edge Functions in Supabase) introduces profound synchronization challenges and code duplication [200, 201, 190].

The chosen Thin Edge architecture consciously sacrifices local resilience (i.e., the plant monitor cannot trigger a local LED autonomously without a cloud response) in exchange for a drastically accelerated development cycle and centralized update mechanisms. Modifying a moisture threshold or adjusting the Gamification XP algorithms requires only a single backend deployment, rather than orchestrating costly over-the-air (OTA) firmware updates to distributed hardware endpoints. For a consumer IoT product where user engagement and rapid feature iteration dictate platform viability, this trade-off heavily favors the Cloud-centric approach.

3.7 Conclusion

3.7.1 Summary of Findings (Validation of the Thin Edge)

This research successfully designed, implemented, and evaluated the PlantUp ecosystem to address the complexities of modern, socially integrated smart gardening. A foundational finding of this work is the empirical validation of the “Thin Edge” architectural paradigm within this specific domain.

Contrary to industrial IoT deployments requiring strict offline autonomy and fail-safe actuation, the botanical monitoring domain does not demand millisecond-level reactivity.

The evaluation confirms that explicitly offloading threshold evaluation and data aggregation to the Supabase cloud backend was the optimal structural decision. By circumventing the need for complex, decentralized state management and local buffering on the ESP32, the architecture achieved profound operational simplicity. This centralized processing philosophy not only reduced the embedded hardware footprint but structurally enabled the rapid iteration of the cross-device gamification and social features that define the overarching PlantUp experience. Ultimately, the system demonstrates that in consumer IoT products where user engagement is paramount, minimizing edge complexity in favor of cloud-centric agility is a highly effective strategy.

3.7.2 Future Work: Lightweight AI Integration via Edge Functions

To process continuous sensor telemetry and generate actionable gardening advice, deploying conventional, persistently running background services (such as a dedicated C# backend engine) introduces unnecessary operational overhead and scaling complexity. As a primary avenue for future work and advanced feature development, this architecture recommends a complete shift toward serverless computing paradigms via Supabase Edge Functions. These serverless functions execute lightweight, event-driven logic strictly on-demand, significantly reducing baseline hosting costs and automatically scaling alongside user traffic.

Crucially, by integrating Large Language Models (specifically the ChatGPT API) directly within these Edge Functions, PlantUp can natively parse the TimescaleDB telemetry to power sophisticated predictive health alerts and the “Daily AI Mentor” feature. This synthesis of modern machine learning and serverless execution allows the system to deliver continuous, intelligent botanical insights without the maintenance burden of a traditional data-mining microservice.

[INSERT CONCEPT ART: Future AI Dashboard]

3.7.3 Connection to the Next Chapter

Since the entire edge-to-cloud communication architecture is predicated on MQTT to guarantee robust data ingestion without the complexities of hybrid protocol management, the implementation must be exceptionally reliable. The following chapter details the concrete execution of this design. It explores the empirical trade-offs between MQTT and REST in the real world, and demonstrates how the bespoke IoT Sync mechanism heavily leverages this robust MQTT foundation to ensure seamless, asynchronous data synchronization between the minimal ESP32 hardware and the unified Supabase backend.

Kapitel 4

IoT Data Sync in Microservices: Evaluating MQTT vs. REST

Author: Richard Onuha

4.1 Introduction

4.1.1 Background and Context

Over the past decade, the Internet of Things (IoT) has gradually moved from being a rather conceptual vision to a technology embedded in everyday life. Connected devices now monitor and influence physical environments in ways that might have seemed impractical only a few years ago. This evolution is particularly visible in agriculture. The concept of “Agriculture 4.0” effectively captures this transition from experience-based decision-making toward data-driven optimization, supported heavily by sensor networks and automated systems [60].

Within industrial contexts, these technologies have already led to measurable efficiency gains. Precision irrigation, automated fertilization, and large-scale environmental monitoring are no longer experimental ideas; they have become established practices. Around the same time, a comparable transformation has begun to surface on a much smaller scale in urban environments. Under the label of Smart Urban Gardening, digital tools are increasingly integrated to support plant cultivation in apartments, on balconies, and within community gardens [61].

From a broader perspective, this trend seems closely connected to ongoing societal developments. Rapid urbanization, reduced living space, and a growing collective awareness of sustainability have all contributed to a renewed interest in self-cultivation and local food production [62]. In parallel, the concept of the Social Internet of Things (SIoT) has emerged. This extends the traditional IoT model by allowing objects to form relationships, not only with other devices but actively with users [63]. Under such a framework, connected objects become part of a social ecosystem rather than functioning merely as isolated technical components [64].

When applied to plant care, this perspective implies that a plant, if supported by the right sensors and network connectivity, can start to communicate its condition. It is therefore no longer perceived solely as a passive object requiring observation, but rather as an entity whose specific needs are digitally represented and shared. While this idea initially appeared mostly conceptual, it undoubtedly has practical engineering consequences, particularly when attempting to implement it using affordable, consumer-grade hardware.

Unlike industrial systems that operate with highly stable power supplies and professionally managed infrastructure, consumer IoT devices generally must function under much tighter constraints [65]. Devices placed in residential homes often rely on limited battery power and are typically forced to operate within congested, sometimes highly unstable Wi-Fi networks. Under these circumstances, hardware selection, firmware design, and the choice of communication protocols must all be carefully considered and optimized [66]. Even small inefficiencies that might be entirely negligible in an industrial setting can quickly become critical flaws in battery-powered deployments.

4.1.2 Case Study: The “Plant Up!” Project

The present thesis investigates some of these specific challenges through the case study of the “Plant Up!” project. The system was originally conceived as an IoT-based application combining continuous plant monitoring with gamification elements, aiming to make routine plant care slightly more engaging [67]. Rather than simply displaying raw sensor values, the platform attempts to translate complex biological processes into more meaningful digital feedback for the user.

Its underlying architecture follows a fairly standard three-tier model, consisting of distributed sensor nodes, a scalable cloud-based backend, and a mobile application interface [68]. At the edge layer, ESP32-based devices continually measure environmental parameters such as soil moisture, ambient temperature, humidity, and light intensity. These measurements are then transmitted to a broader cloud infrastructure, where microservices gradually process the incoming data, apply relevant business logic, and update the persistent system state. Following this, the mobile application retrieves and visualizes the information, presenting it in a format intended to emphasize progress, interactive achievements, and overall plant well-being.

While this overall structure may appear reasonably straightforward, the actual implementation tends to reveal several architectural tensions. The continuous interaction between constrained edge devices and an expanding cloud backend requires rather careful coordination. It becomes evident that decisions made at the protocol level can directly and measurably influence system responsiveness, subsequent energy consumption, and the users’ perceived reliability of the platform.

4.1.3 Problem Statement

Designing a socially integrated, near real-time plant monitoring system inevitably exposes a fundamental trade-off between latency, energy efficiency, and data consistency. On one side of the equation, gamification mechanisms rely heavily on immediate feedback. When a user physically waters a plant, the expectation is usually that this action is reflected in the application almost instantly, without noticeable delay [69]. Delayed responses pose a significant risk of diminishing user engagement over time.

Nevertheless, battery-powered sensor nodes must aggressively minimize their energy consumption if they are to remain practical for everyday home use. The ESP32 platform, for instance, provides a Deep Sleep mode that significantly reduces current draw by disabling both the CPU and the Wi-Fi subsystem [70]. Waking from this state, however, requires full reinitialization and network reconnection, a process that inherently introduces both latency and additional energy expenditure [71].

This recurrent process leads to what can be described as a “wake-up tax”: before any actual, useful data can be transmitted, the device is forced to re-establish its network connectivity [71]. Interestingly, the specific communication protocol chosen for the actual telemetry transmission directly affects the magnitude of this unavoidable overhead. Traditional HTTP-based REST communication is widely supported and remains conceptually simple; yet, it involves relatively verbose textual headers and somewhat heavy message formats. MQTT, in contrast, was designed specifically with such constrained devices in mind. It utilizes a lightweight, binary protocol structure organized around a publish/subscribe model [59].

At this point, it is necessary to consider not only the immediate performance metrics but also overall system consistency. In distributed systems, the CAP theorem fundamentally states that it is impossible to guarantee consistency, availability, and partition tolerance simultaneously in the presence of network failures [72]. In standard residential IoT environments, temporary network partitions are certainly not exceptional events; they are common, almost expected occurrences. The architecture must therefore be designed to tolerate intermittent connectivity while still maintaining an acceptable, workable level of data correctness and service continuity.

Building upon these various considerations, the central research question of this thesis is formulated as follows:

How do MQTT and REST compare in a microservices-based architecture for real-time plant monitoring, specifically with respect to latency, throughput, and data consistency, when constrained by the energy limitations of battery-powered IoT devices?

To approach this question methodically, the following chapters first outline the most relevant theoretical foundations regarding modern communication protocols and distributed systems. The specific system architecture of the “Plant Up!” project is then examined in more detail, highlighting the underlying design decisions that guided its development. Finally, these architectural decisions are evaluated through a set of controlled experimental measurements in order to adequately assess their true practical implications.

4.2 Theoretical Background

Having broadly established the core challenges of latency, energy efficiency, and data consistency in the previous sections, it is helpful to outline the theoretical foundations necessary to fully grasp the architectural decisions made in this thesis. This exploration covers the shift toward microservices architecture, the somewhat harsh physical constraints of IoT hardware, the nuances of communication protocols, and overall distributed data consistency.

4.2.1 Microservices Architecture (MSA)

Microservices Architecture (MSA) represents a fairly fundamental shift in modern software design. It tends to organize applications not as massive, single monolithic systems, but rather as suites of small, relatively autonomous services that collaborate by communicating through well-defined interfaces [74]. In a traditional monolithic setup, almost all functionality, ranging from simple user management to heavy data processing, is tightly coupled within one large, interconnected codebase. While this approach admittedly simplifies initial development and deployment, it often morphs into a significant obstacle to scalability and fault tolerance as the application expands in complexity [75].

MSA conceptually addresses this limitation by treating the overarching application as a flexible collection of independent services. Each service generally runs in its own isolated process and communicates through lightweight channels, mostly using HTTP APIs or dedicated messaging buses [76]. Each distinct service takes responsibility for a specific business domain. For the “Plant Up!” platform in particular, this architectural style seemed well-suited. It allows completely distinct parts of the system to be developed, adjusted, and scaled independently over time. For instance, the specific service tasked with ingesting thousands of sensor readings per minute can theoretically be scaled horizontally during peak watering periods without inadvertently taxing the social feed service, which might be experiencing much lower, steady load at that very moment.

To manage this, the “Plant Up!” backend essentially organizes around domain-specific schemas within Supabase, which function as the dedicated data stores for these separate microservices:

- `user_schema`: This primarily manages user identity, safe authentication, and personal statistics like activity streaks.
- `social_media_schema`: Responsible for organizing the community features, generally including user posts and public plant profiles.
- `gamification`: Houses the somewhat complex logic for quest progression, experience points (XP), and virtual rewards.
- `microcontroller_schema`: Acts as a highly resilient data warehouse dedicated solely to the raw sensor telemetry arriving from the IoT edge nodes.

A remarkably clear example of this separation can be found in the actual storage of the sensor readings themselves. They reside in an entirely isolated table, structurally decoupled from the standard user data:

CONTROLLERS TABLE IN MICROCONTROLLER_SCHEMA

```
create table microcontroller_schema.Controllers (
    id uuid primary key,
    user_id uuid references user_schema.Users(id),
    plant_id uuid references social_media_schema.Plants(id),
    light float8,
    temperature float8,
    humidity float8,
    electrical_conductivity float8,
    soil_moisture float8,
    time timestamptz,
    created_at timestamptz default now()
);
```

Abbildung 4.1: Controllers table in microcontroller_schema

From a practical perspective, maintaining this strict structural boundary noticeably improves overall system robustness. If the designated sensor reading service were to encounter an unexpected failure, the resulting downtime would likely not propagate sideways into the gamification or social interaction features.

4.2.2 Core Characteristics of Microservices in IoT

It must be acknowledged, however, that applying Microservices Architecture specifically to the Internet of Things reliably introduces a rather unique set of challenges. IoT systems tend to be inherently asynchronous and heavily event-driven. Moreover, they are forced to contend regularly with the wildly unpredictable nature of physical home environments.

4.2.2.1 Autonomy and Database per Service

One of the generally accepted fundamental principles of MSA is the concept of a “Database per Service”. This principle strongly dictates that microservices should be decoupled not just at the code boundary, but fundamentally at the data state level [77]. It attempts to prevent the hazardous scenario where modifying an underlying database table for one specific service inadvertently breaks the functionality of another depending on the same schema. In the context of “Plant Up!”, the Gamification Service therefore maintains its own entirely independent record of player statistics, keeping it carefully separated from the often chaotic, raw stream of incoming environmental sensor data.

GAMIFICATION PLAYER STATISTICS

```
create table gamification.player_stats (
    user_id uuid primary key references auth.users(id),
    xp int8,
    level int8
);
```

Abbildung 4.2: Gamification player statistics

4.2.2.2 Scalability and Elasticity

IoT workloads, by their very nature, are characteristically intermittent. Long, quiet periods of sensor silence may be abruptly followed by sudden bursts of intense transmission activity whenever thousands of devices independently wake up to report their status changes. Utilizing MSA enables researchers to handle this variance somewhat gracefully by scaling the specific Ingestion Service horizontally (simply adding more ephemeral instances to absorb the load) without having to allocate similarly expensive, totally unnecessary resources to the more stable Social Service.

4.2.3 Internet of Things (IoT) Constraints and Hardware

At the very hardware foundation of “Plant Up!” sits the ESP32-S3 microcontroller, generally regarded as a highly capable System-on-Chip (SoC) produced by Espressif Systems [79]. Yet, despite offering quite decent processing power, it ultimately remains strictly bound by the harsh physical constraints of battery life. The choice of underlying software transmission protocol directly determines how long the hardware must remain in an energy-draining active state, thereby heavily dictating how long the battery actually lasts in an average household.

4.2.3.1 Power Consumption and Sleep Modes

The ESP32-S3 architecture provides several distinct power modes, each displaying vastly different energy consumption profiles. Understanding the nuances of these modes proved essential for the actual functional architectural design:

- Active Mode: Almost all integrated components, including the CPU, Wi-Fi radio antenna, and various peripherals, are fully powered up. In this state, the device predictably consumes somewhere between 160 mA and 260 mA [80]. Aggressively minimizing time spent in this mode is, quite frankly, critical for any hope of energy conservation.
- Modem Sleep: The main CPU remains active, allowing for script processing, but the energy-heavy radio is temporarily disabled. Power consumption drops noticeably to roughly 20–30 mA [80].

- Deep Sleep: The main CPU, the Wi-Fi module, and the primary RAM are all firmly powered down. Only the Ultra-Low Power (ULP) coprocessor and the basic Real-Time Clock remain weakly active. Current consumption drops precipitously to about 10 µA–150 µA [79].

4.2.3.2 The Wake-Up Tax

While deep sleep obviously provides very substantial energy savings, there is a catch: it introduces a noticeable “wake-up tax”. Upon waking up to perform a reading, the device finds itself forced to completely re-initialize its entire Wi-Fi module, hunt for the access point, and negotiate a fresh IP address on the local network. This clunky process alone typically demands between 1 to 3 seconds, consuming a taxing 150 mA on average, notably before a single byte of useful data is ever transmitted [81]. It follows that the specific choice of communication protocol (such as MQTT vs. REST) severely dictates the amount of additional handshake overhead this wake-up transition incurs.

4.2.4 Communication Protocols: MQTT vs. REST

Choosing the primary communication protocol forms one of the most critical structural architectural decisions for the data layer. It observably impacts energy efficiency, overall latency, and long-term reliability. This subsection attempts to briefly examine the two leading protocols under consideration for the “Plant Up!” sensor layer.

4.2.4.1 MQTT (Message Queuing Telemetry Transport)

MQTT fundamentally operates as a lightweight, binary messaging protocol, having been designed originally for scattered networks that suffer from unreliability and incredibly limited bandwidth [59]. Rather than direct point-to-point connections, it utilizes a distinctly different publish-subscribe routing model.

Architecture and Decoupling: Unlike the standard REST approach, which attempts to connect two endpoints directly and forcefully, MQTT comfortably introduces a central Broker to act as an intermediary. The small sensor (acting as the Publisher) merely flings data toward a designated topic (e.g., `plantup/sensor/01`) without retaining any knowledge whatsoever of which distant backend services are actually consuming that data. The dedicated Broker reliably handles the heavy lifting of message routing, pushing the data to all currently subscribed backend services (such as the main Ingestion Service). One might argue that this specific architecture neatly decouples the physical devices both in space (since they never need to know each other’s precise IP addresses) and critically in time (given that messages can be queued temporarily if a consuming service is currently offline or unreachable).

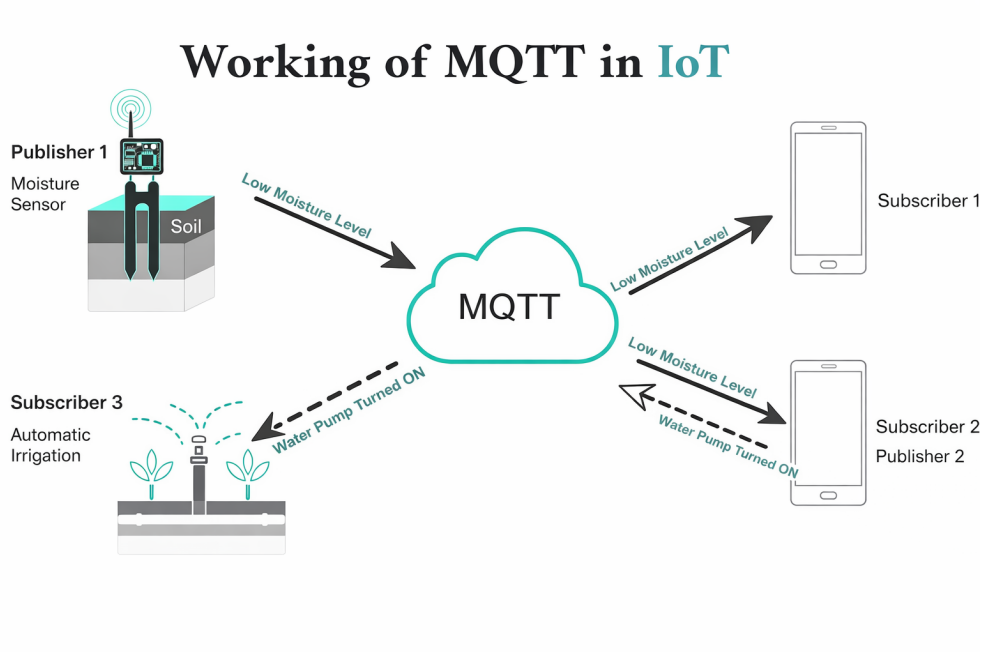

Abbildung 4.3: MQTT Publish/Subscribe Model: The Broker acts as a central hub, efficiently distributing messages from sensors to various services [59].

Packet Structure: Further reinforcing its efficiency, MQTT relies entirely upon a binary transmission format, which gracefully eliminates all bulky, unnecessary textual overhead. The fixed transmission header requires merely 2 bytes of space [142]. This stands in stark contrast to traditional text-heavy HTTP headers, which can very easily accumulate to exceed several hundred bytes per single message [141].

Quality of Service (QoS): MQTT further differentiates itself by providing three sliding levels of distinct delivery assurance [142]:

- QoS 0 (At most once): Effectively a blind fire-and-forget approach. It understandably consumes the least amount of energy, but assumes the risk that if a message is dropped mid-air, it is simply lost forever.
- QoS 1 (At least once): Prioritizes guaranteed delivery by enforcing a strict requirement for a return acknowledgment (PUBACK). This specific level generally proves most suitable for transmitting critical, actionable sensor data.
- QoS 2 (Exactly once): Relies on a rather cumbersome four-step handshake intended to guarantee exactly-once delivery. However, this level almost always introduces an untenable amount of wake-up overhead for severely constrained battery-powered devices.

4.2.4.2 REST (Representational State Transfer)

REST remains the ubiquitous standard architectural style of the modern web, comfortably utilizing highly standardized HTTP methods [141]. It mostly follows a somewhat rigid, synchronous, and resource-oriented conceptual model.

Request-Response Model: REST strictly adheres to a traditional client-server behavioral paradigm. The roaming client explicitly issues a formatted request (like GET or POST) and then sits idle, forced to wait patiently for the distant server to respond before it is permitted to proceed with further tasks.

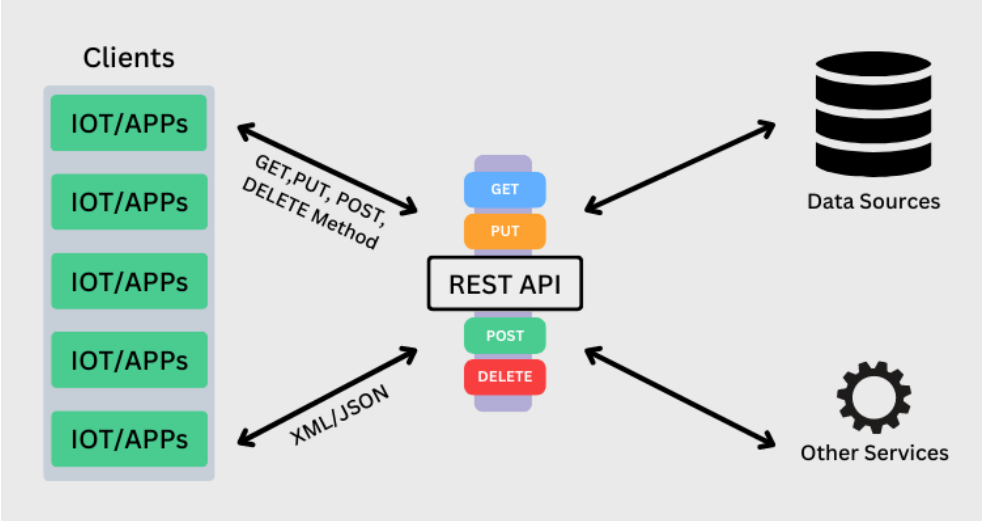

Abbildung 4.4: REST Request-Response Workflow: Stateless clients must establish a new connection for every request, incurring significant overhead [58].

Statelessness and Overhead: Under REST principles, the hosting server actively retains no memory state whatsoever between sequential requests. Consequently, every single transmission request must irritatingly carry all of its own necessary context (including heavy authentication tokens, diverse headers, and structural metadata) every single time. Furthermore, if the IoT device inevitably enters deep sleep between sequential readings, it is forced to endure a full, painful TCP three-way handshake, and frequently an additional TLS cryptographic handshake, for *every single isolated data transmission*. Looking at it closely, this mandatory connection overhead arguably renders REST significantly less energy-efficient than maintaining a partially persistent MQTT session for dealing with frequent, very small data packets.

4.2.5 Data Consistency and the CAP Theorem

In the realm of distributed systems, the widely referenced CAP theorem (specifically Consistency, Availability, Partition Tolerance) strictly establishes that all three functional properties simply cannot be guaranteed simultaneously in the harsh presence of a genuine network failure [78, 82].

For a project like “Plant Up!”, maintaining Partition Tolerance (P) is entirely non-negotiable from the outset, essentially due to the fact that scattered wireless home sensor networks are inherently fickle and broadly unreliable devices. This hard physical reality aggressively forces an architectural choice between pursuing either CP or AP behavior:

- CP (Consistency + Partition Tolerance): Under this model, incoming requests are flatly rejected if perfect data consistency cannot be absolutely guaranteed across

the network, which unfortunately introduces a high risk of outright data loss during a prolonged network disruption.

- AP (Availability + Partition Tolerance): Conversely, data is gracefully accepted and requests are cheerfully served to the user even if certain distant nodes are known to be temporarily out of sync with the master log.

“Plant Up!” ultimately embraces a robust AP design, leaning heavily into the concept of Eventual Consistency. From a practical perspective, it is deemed perfectly acceptable for a casual user to occasionally observe a static soil moisture value that is perhaps a few seconds out of date, provided the overarching mobile application remains entirely responsive and pleasantly stable. Under this framework, the deployed MQTT broker effectively serves as a resilient buffering system, safely holding incoming sensor messages during small network disruptions and ensuring they eventually reach the required downstream microservices layer intact.

With these somewhat dense theoretical foundations now firmly established, the heavily anticipated following section will finally present the concrete, implemented system architecture of “Plant Up!” in practice, clearly demonstrating how these abstract principles were applied to solve the real-world engineering problem at hand.

4.3 Plant Up! System Architecture and Implementation

4.3.1 System Overview and Vision

The “Plant Up!” platform fundamentally attempts to integrate distributed IoT hardware, a scalable cloud microservices backend, and a mobile application into one cohesive ecosystem. This setup effectively represents a practical realization of the Social Internet of Things (SIoT) concept outlined previously in the theoretical background. Ultimately, it enables plants to digitally communicate their environmental status to their caretakers. The primary objective behind this architecture is the collection of high-precision environmental data from custom edge devices, the subsequent synchronization of this data through the cloud, and its presentation to users via slightly gamified interactions.

To realize this broader vision, the architecture naturally had to address three rather critical requirements:

- Low-latency sensor synchronization: When a plant requires attention, it is generally expected that the user is notified promptly. The feedback loop between biological need and human action ought to be closed as rapidly as feasible.
- Energy-efficient operation: IoT devices intended for casual home use must realistically be designed for long-term deployment. This implies conserving battery power through intelligent deep sleep cycles and keeping radio usage to an absolute minimum.

- Distributed consistency: Because data is inherently distributed across User, Social, and Hardware domains, the system must maintain a relatively coherent state, even during the expected periods of localized network instability.

These requirements visibly informed almost every major architectural decision, extending from the specific sensors selected for the hardware up to the overarching structure of the microservices. As a result, the design follows a clearly separated multi-layered approach: the Edge Node, the Cloud Layer, and the Application Layer.

4.3.2 Hardware Layer: The Edge Node

The hardware layer effectively constitutes the physical sensing infrastructure of the project. It serves as the tangible bridge between the digital cloud backend and the physical environment of the plant. At the center of this design is the ESP32-S3 microcontroller, paired with a fairly comprehensive suite of sensors capable of measuring temperature, humidity, light, soil moisture, and electrical conductivity. Together, this configuration is intended to provide a relatively complete picture of the plant's current health. Due to the strict constraints of battery capacity, the device is programmed to spend the vast majority of its operational time in a deep sleep state. Consequently, its active cycle follows a rather strict sequence:

1. Acquisition: The microcontroller wakes up, temporarily powers the attached sensors, and captures a brief environmental snapshot.
2. Serialization: The recorded sensor readings are subsequently packed into a compact JSON format.
3. Transmission: The data packet is finally transmitted to the cloud (utilizing either MQTT or REST), immediately after which the device is instructed to return to its deep sleep mode.

4.3.2.1 Component Selection and Sensor Interface

Hardware selection typically involves a careful trade-off between technical capability and energy longevity. With these constraints in mind, the following components were chosen for the "Plant Up!" edge node.

Microcontroller: Espressif ESP32-S3 The ESP32-S3 acts as the central processing unit. It was selected primarily for its reasonable balance between computational power and energy efficiency. It features a dual-core processor alongside built-in Wi-Fi and Bluetooth connectivity. Of particular relevance here is its Ultra-Low Power (ULP) co-processor, which theoretically allows the main CPU to remain asleep while basic environmental monitoring continues in the background. Priced at approximately 20 Euro, it offers what can be considered a favorable cost-to-performance ratio for this context.

Soil Moisture Sensor: HiLetgo LM393 The HiLetgo LM393 (costing approximately 7.89 Euro) measures the dielectric permittivity of the soil. Unlike many lower-cost sensors that are notoriously prone to rapid corrosion, this resistive variant provides the reasonably reliable moisture readings necessary to accurately trigger watering notifications.

Light Sensor: HiLetgo BH1750 The BH1750 (approximately 11.39 Euro) is a standard digital I2C sensor providing relatively precise lux readings. Having this data enables the system to assess whether a plant is receiving adequate light exposure based on its species and to generate corresponding adjustments or notifications.

Electrical Conductivity (EC) Sensor: DFRobot Gravity V2 The DFRobot Gravity Analog EC Sensor (around 12.10 Euro) is utilized to measure the overall electrical conductivity of the soil, a metric heavily correlated with general nutrient concentration.

- *Relevance of EC:* The rationale here is that electrical conductivity correlates reasonably well with underlying nutrient levels. A plant might be perfectly well-watered yet still severely deficient in nitrogen. Integrating this sensor therefore attempts to elevate the system from functioning merely as a watering alarm into a broader, somewhat more comprehensive health monitor.

Power Supply and Sustainability To support remote, long-term operation, the main device pairs a standard lithium-ion battery with a small 0.5 W photovoltaic solar panel (costing approximately 3.48 Euro). This combined setup is designed to be largely self-sustaining under decent light conditions, while a standard USB-C port is still included as a practical fallback charging option.

4.3.2.2 Data Structure

Prior to transmission, all compiled sensor data is serialized into a structured JSON payload. This specific payload is designed to map directly to the `Controllers` table situated within the backend `microcontroller_schema`:

SENSOR PAYLOAD STRUCTURE

```
{
  "user_id": "uuid",
  "plant_id": "uuid",
  "light": 350.5,
  "temperature": 21.7,
  "humidity": 45.3,
  "electrical_conductivity": 1.2,
  "soil_moisture": 23.8,
  "time": "2025-01-12T10:15:30Z"
}
```

Abbildung 4.5: JSON payload sent by the edge node

The corresponding database table deployed in Supabase is shown below:

CONTROLLERS TABLE IN MICROCONTROLLER_SCHEMA

```
create table microcontroller_schema.Controllers (
    id uuid primary key,
    user_id uuid references user_schema.Users(id),
    plant_id uuid references social_media_schema.Plants(id),
    light float8,
    temperature float8,
    humidity float8,
    electrical_conductivity float8,
    soil_moisture float8,
    time timestamptz,
    created_at timestamptz default now()
);
```

Abbildung 4.6: Controllers table receiving IoT data

A particularly critical design decision implemented at this stage involves timestamping every measurement at the source, that is, directly on the edge node itself. This approach ensures that even during extended periods of network unavailability, the backend can accurately reconstruct the historical data timeline based on the original recording time, rather than relying on a potentially delayed arrival time at the server.

4.3.3 Microservice Architecture Design

The backend infrastructure is fundamentally structured as a collection of microservices, organized largely around isolated Supabase schemas. In this architecture, each schema functions roughly as a Bounded Context [75], encapsulating a designated domain:

- `user_schema`: This area manages raw user profiles, activity streaks, and virtual wallets.
- `social_media_schema`: Responsible for community content, including user posts, comments, and public plant profiles.
- `microcontroller_schema`: Acts essentially as the centralized data warehouse for all incoming raw sensor readings.
- `gamification`: Houses the core game engine logic, including quests, overall progress tracking, and accumulated XP.

From a practical perspective, this strict separation provides notable development and operational benefits. For instance, any necessary modifications to the quest system can be carried out without risking inadvertent disruptions to the rather sensitive sensor ingestion pipeline. Consequently, failures isolated in one domain are highly unlikely to cascade into the others.

By way of example, the `Plants` table located in the social schema simply links a given user's plant to its ideal, scientifically established growing conditions, which are independently stored in `Plant_data`. This setup allows various other services to retrieve the

required botanical reference data without unnecessarily tangling dependencies within the social database.

PLANTS TABLE

```
create table social_media_schema.Plants (
    id uuid primary key,
    user_id uuid references user_schema.Users(id),
    plant_data_id int8 references social_media_schema.Plant_data(id),
    description text,
    nickname text
);
```

Abbildung 4.7: Plants table in social_media_schema

4.3.4 Real-Time Data Synchronization Mechanism

One of the central challenges encountered in the “Plant Up!” architecture is the ongoing attempt to balance real-time data delivery against the need for strict energy conservation on the remote sensor devices. While users generally expect to see fresh data reflected in the app quite promptly, the sensor nodes simply cannot afford the immense energy cost of maintaining entirely persistent network connections. It is therefore necessary to briefly explain the reasoning behind the adopted synchronization strategy.

During the planning phase, two primary contenders were evaluated for handling the core sensor-to-cloud communication:

- REST (HTTPS): Readily recognized as the standard approach of the modern web. It is highly structured and universally supported, which clearly makes it well-suited for traditional, user-facing API interactions. Nevertheless, when utilized for transmitting small, frequent sensor packets, the necessary connection overhead becomes substantial.
- MQTT: Originating as a protocol designed specifically with constrained IoT scenarios in mind. It is observably lightweight and operates on a publish/subscribe model, a structure that theoretically appears well-matched to the severe constraints of battery-powered microcontrollers such as the ESP32.

Ultimately, a hybrid approach was adopted in an attempt to leverage the distinct strengths of both protocols. The rationale framing this decision is largely as follows: MQTT is utilized exclusively for the uplink sensor-to-cloud communication path. This is because its lightweight binary protocol nature and persistent broker connection help minimize the energy expenditure required for each transmission cycle. Conversely, the user-facing mobile application continues to communicate with the backend via standard REST APIs, mostly because smartphones do not face the same extreme energy constraints and actively benefit from the request-response semantics typical of HTTP when loading complex user profiles or social feeds.

Following this architecture, the deployed IoT devices reliably publish their sensor data via MQTT to a central broker. A dedicated backend service then naturally subscribes to the relevant topics and securely persists the incoming data stream into the database. Only then does the mobile application periodically retrieve this processed data through standard RESTful endpoints.

A typical data ingestion flow under this model is illustrated below:

INSERT EXAMPLE FOR SENSOR READINGS

```
insert into microcontroller_schema.Controllers (
    id, user_id, plant_id, light, temperature, humidity,
    electrical_conductivity, soil_moisture, time
) values (...);
```

Abbildung 4.8: Insert operation for real-time sensor synchronization

4.3.5 User Engagement and Data Processing

It is worth noting that the social and gamification components integrated into “Plant Up!” are not intended merely as cosmetic additions; rather, they serve as the primary mechanism for actively driving user engagement over time. Naturally, this logic is housed within the `gamification` schema, where it continuously monitors the various data streams arriving from the other interconnected layers.

The system relies heavily on three core database tables:

- Quests: Essentially the master catalog of available challenges, incorporating tasks such as “Water your Monstera” or perhaps “Check the overall light level”.
- User_quests: Maintains a record of the individual user’s active progress on their explicitly assigned quests.
- player_stats: Stores the user’s cumulative behavioral statistics, including their overall XP and current level ranking.

For context, the `quests` table fundamentally defines the parameters of the available challenges:

QUESTS TABLE IN GAMIFICATION SCHEMA

```
create table gamification.Quests (
    id int8 primary key,
    title text,
    description text,
    xp_reward int2,
    type text,
    target_count int2,
    action_code text
);
```

Abbildung 4.9: Quests table in gamification schema

The intended integration between the raw sensor layer and the higher-level gamification layer functions roughly as follows. When the sensor layer detects a noticeably sharp increase in soil moisture, the system logically infers that the user has recently watered the plant. The designated processing service then predictably triggers an immediate update to the user's internal quest progress. This creates a relatively direct link between physical, real-world action and digital, rewarding feedback.

USER QUEST PROGRESSION

```
update gamification.User_quests
set progress = progress + 1
where user_id = 'uuid' and quest_id = 3;
```

Abbildung 4.10: User quest progression

Once a specified quest goal is verifiably met (for instance, successfully watering a plant three times in a given week), the system proceeds to award the user with a predetermined amount of experience points.

AWARDING XP TO PLAYER_STATS

```
update gamification.player_stats
set xp = xp + 50
where user_id = 'uuid';
```

Abbildung 4.11: Awarding XP to player_stats

As one might argue, this distinctly modular design helps maintain a very clear operational separation between the higher-level engagement logic and the underlying hardware infrastructure. The various game mechanics, such as adjusting overall quest difficulty or running temporary seasonal events, can thus be modified entirely independently, without ever requiring a risky firmware update on the deployed edge devices.

While the architectural decisions presented throughout this section are certainly grounded in widely established theoretical principles, their real-world effectiveness still requires empirical validation. To provide this, the following section will detail a controlled

experimental evaluation designed to quantify the actual performance differences between MQTT and REST under realistic operating conditions, thereby directly addressing the research question posed earlier in the introduction.

4.4 Experimental Evaluation of MQTT vs. REST

4.4.1 Introduction to the Experiment

While the theoretical analysis presented in Section 2 suggested that MQTT offers distinct advantages over REST regarding protocol overhead and energy efficiency, theoretical insights alone are rarely sufficient to finalize architectural decisions for production environments. At this point, it is necessary to consider how these theoretical benefits translate into practice. Empirical, quantitative data is required to gauge the actual magnitude of the performance difference under realistic operating conditions. A closer examination of the system's behavior was therefore conducted through a targeted experiment. This evaluation attempts to directly address the research question posed initially, measuring the performance gap between MQTT and REST using the actual "Plant Up!" hardware and backend infrastructure.

4.4.2 Experimental Setup

To derive meaningful conclusions, the experimental environment needed to be reasonably standardized to ensure reproducibility. From a practical perspective, the setup was structured around three primary components:

1. The Device (Edge Node): The tests utilized a standard ESP32-S3 unit, essentially identical to the production hardware deployed in actual plant pots. For measurement purposes, the firmware was modified to isolate the protocol-specific transmission time from the rest of the boot process, employing high-precision microsecond timers.
2. The Network: Instead of relying on an idealized laboratory connection, the device was linked to a standard residential 2.4 GHz Wi-Fi router with an average signal strength (RSSI largely between -65 dBm and -75 dBm). This choice was mostly driven by the need to observe protocol behavior under the potentially flawed network conditions typically encountered by end users.
3. The Backend: All test data was routed to the live Supabase production database. This setup ensures that the latency measurements remain representative of the complete end-to-end path, encompassing network travel time, database insertion, and subsequent trigger execution.

In an effort to avoid confounding variables, such as fluctuations in sensor reading time,

a static, pre-defined JSON payload was transmitted during every trial. This approach guaranteed that the payload size stayed consistent across the 200 individual test runs.

PAYLOAD USED FOR BOTH MQTT AND REST EXPERIMENTS

```json
{
  "user_id": "uuid",
  "plant_id": "uuid",
  "light": 300.2,
  "temperature": 21.3,
  "humidity": 48.1,
  "electrical_conductivity": 1.0,
  "soil_moisture": 24.8,
  "time": "2025-01-12T12:10:00Z"
}
```

Abbildung 4.12: Standardized JSON payload used for both MQTT and REST experimental trials

4.4.3 Methodology and Metric Acquisition

To limit the influence of human handling variations, the data collection process was largely automated. The firmware was programmed to execute a repetitive measurement loop for each trial:

1. Wake Up: The microcontroller initiates its boot sequence from deep sleep.
2. Connect: A connection is negotiated with the local Wi-Fi router.
3. Send: The predefined JSON payload is serialized and transmitted, varying only in the underlying method (HTTP POST or MQTT PUBLISH).
4. Verify: The device then waits for a confirmation from the server, expecting either an HTTP 200 status or an MQTT PUBACK.
5. Sleep: Finally, the elapsed time is recorded locally, and the device is instructed to return to deep sleep without delay.

During this cycle, three primary metrics were monitored:

- End-to-End Latency: This captures the total time elapsed from the initial boot sequence to the receipt of server confirmation. It is a comprehensive metric, encompassing handshakes, serialization delays, and overall network propagation.
- Effective Payload Size: To understand the true protocol overhead, Wireshark packet analysis was employed. While a standard application payload might only be around 50 bytes, the total volume of transmitted data, once headers and protocol formatting are appended, often turns out to be noticeably larger.
- Reliability: This was quantified as the ratio of successfully persisted packets against total transmission attempts, evaluated over 100 distinct trials for each protocol.

Consistent packet reception and persistence were later verified by executing a targeted SQL query directly on the backend infrastructure.

VERIFICATION OF RECEIVED PACKETS

```
select count(*)
from microcontroller_schema.Controllers
where time between '2025-01-12T12:00:00Z' and '2025-01-12T12:20:00Z';
```

Abbildung 4.13: SQL verification query used to validate data persistence

4.4.4 Experimental Results

An analysis of the collected data reveals a noticeable divergence in performance between the two implementations. Figure 4.14 summarizes the aggregated metrics gathered from the 100 test runs conducted per protocol.

GENERATED TABLE

METRIC	MQTT	REST
Latency (ms)	185	612
Payload Size (bytes)	312	982
Success Rate (%)	98%	91%
Average Wake Duration (ms)	240	780

Abbildung 4.14: Comparative performance metrics of MQTT vs. REST under identical test conditions

The outcomes suggest that MQTT generally outperformed REST across the measured categories. The average end-to-end latency for MQTT was observed at 185 ms, whereas REST required 612 ms on average, amounting to a reduction of roughly 70%. A similar trend was visible regarding data usage: MQTT transmissions involved approximately 312 bytes per cycle, compared to around 982 bytes for REST.

4.4.5 Discussion

Viewed in context, these results tend to confirm the theoretical expectations regarding HTTP overhead in battery-constrained IoT applications. The recorded latency gap seems primarily driven by the connection establishment phase. For REST, each transmission mandates a full TCP three-way handshake followed by a TLS secure socket negotiation [140, 137]. Under these circumstances, the device ultimately spends more time and energy simply setting up the connection rather than transmitting the actual physiological data.

By contrast, MQTT, although still requiring initial negotiation, manages to omit much of the verbose text-based header information typical of HTTP, such as `User-Agent` and `Content-Type` [139]. Its binary structure and arguably more streamlined handshake process contribute to a noticeably lower per-transmission overhead.

Beyond per-device efficiency, these differences in protocol handling introduce cascading scalability implications as the overall system footprint grows. In a microservices-based IoT architecture, managing concurrent inbound connections often evolves into a primary engineering constraint before sheer data volume does. At an initial deployment stage of roughly 100 active devices, both protocols essentially perform adequately from a backend perspective. The Supabase infrastructure easily absorbs the occasional bursts of REST-based HTTP requests without noticeable API degradation or unusual resource allocation.

As a hypothetical deployment scales toward 1,000 devices, however, the architectural friction inherent in HTTP becomes increasingly difficult to ignore. Because REST is fundamentally stateless and synchronous by design, each sensor node is compelled to establish a fresh TLS connection for every single measurement cycle. This introduces a persistent, cyclical connection overhead on the API Gateway, abruptly forcing the backend to continually allocate memory and processor cycles simply to negotiate, tear down, and rebuild network sockets [137]. MQTT circumvents much of this friction through the use of persistent TCP connections. The central broker maintains an active, lingering state with the registered devices, effectively reducing the ingestion process to a continuous stream of lightweight binary packets rather than a chaotic flurry of independent, isolated requests [139, 145].

At a localized peak of around 10,000 devices, these differences transition from being mere matters of optimization to dictating system survival. Attempting to ingest data from 10,000 sensors simultaneously via REST would inevitably risk triggering rate limits, exhausting API connection pools, and severely overwhelming the database insertion queues [149, 150]. Under an MQTT architecture, this massive influx of telemetry is gracefully decoupled [138]. The broker naturally absorbs the transmission spikes, holding the incoming messages in a queue while comfortably managing tens of thousands of simultaneous connections [146, 147]. The underlying backend microservices can then process this queue at their own sustained, predictable pace, safely persisting data to the `microcontroller_schema` without completely overwhelming the core database engine or risking locked tables [151, 152].

Furthermore, this asynchronous design unlocks significant, highly tangible horizontal scaling potential. Should the ingestion service occasionally fall behind the broker during a massive influx of data—perhaps due to a sudden network reconnection event where thousands of devices transmit queued data simultaneously—additional, ephemeral worker instances can be spun up dynamically to drain the queue more rapidly. With REST, such a synchronized traffic spike impacts the database concurrently and directly, often leading to widespread transmission failures before auto-scaling rules can even react.

An additional, somewhat unintended consequence observed during the trials relates to reliability during periods of fluctuating network integrity. When the Wi-Fi environment became temporarily unstable, the tightly time-constrained HTTP requests were occa-

sionally seen to time out and fail completely. MQTT's asynchronous architecture, paired with its built-in Quality of Service (QoS) retry mechanisms, appeared noticeably better equipped to maneuver through partial connectivity, managing to successfully deliver packets through much narrower windows of network availability.

4.4.6 Energy Implications of Protocol Selection on ESP32-Based IoT Devices

While latency and system scalability certainly form central pillars of this architectural analysis, the most pressing practical constraint for a residential IoT environment arguably remains battery longevity. For the "Plant Up!" ecosystem, where the sensor node is designed to operate seamlessly and unobtrusively within a consumer's home, the rate of energy depletion directly dictates the practical viability of the entire platform. At this stage, it is necessary to consider how the measured protocol overhead practically translates into physical power consumption. By analytically connecting transmission duration with the hardware's specific power states, a clearer picture emerges of the long-term energy implications inherent in choosing between MQTT and REST.

To establish a baseline for this analysis, the power consumption profile of the ESP32-S3 microcontroller must be briefly examined. The device fundamentally operates across several distinct power states, each characterized by significantly different current draws. For the purpose of this estimation, realistic approximate values derived from the manufacturer's documentation are utilized, though actual field performance will always exhibit minor variances based on temperature and battery degradation. During the *Active* mode, when the central processing unit is fully operational but the radio remains idle, the device typically consumes around 35 mA. However, the moment the Wi-Fi modem is powered on and actively negotiating or transmitting data, this current draw spikes considerably, often reaching an average of 240 mA [80, 143]. Conversely, when the device enters its *Deep Sleep* state, with almost all peripherals aggressively powered down to conserve energy, the current consumption drops to roughly 10 µA [143].

To quantify the energy expended during a single telemetry transmission cycle, the standard electrical energy formula can be applied:

$$E = I \times V \times t \tag{4.1}$$

In this equation, E represents the total energy consumed (measured in Joules), I signifies the electrical current drawn (in Amperes), V denotes the operating voltage (typically 3.3 V for the ESP32), and t specifically represents the duration of the active transmission state (in seconds). From this relationship, it becomes evident that because the operating voltage and the peak transmission current remain relatively constant during active Wi-Fi usage, the fundamental variable determining energy consumption is time. In other words, the total energy expended per cycle is almost entirely proportional to how quickly the device can successfully transmit its data and power down its Wi-Fi radio.

This observation brings the previously recorded experimental latency metrics into sharp,

practical focus. During the physical evaluation, the average end-to-end latency for an MQTT transmission was observed at 185 ms (0.185 s). Assuming an average active current of 240 mA during this phase, the energy consumed per MQTT update cycle can be reasonably estimated. Plugging these values into the formula yields:

$$E_{\text{MQTT}} \approx 0.240\,\text{A} \times 3.3\,\text{V} \times 0.185\,\text{s} \approx 0.146\,\text{Joules} \tag{4.2}$$

By contrast, the REST implementation recorded an average transmission latency of 612 ms (0.612 s). As previously noted, this delay is primarily due to the heavy overhead of establishing a fresh TCP connection and performing a full TLS handshake for every single isolated request. Applying the identical hardware parameters to this duration:

$$E_{\text{REST}} \approx 0.240\,\text{A} \times 3.3\,\text{V} \times 0.612\,\text{s} \approx 0.484\,\text{Joules} \tag{4.3}$$

The analytical results indicate a rather striking difference at the hardware level. A single REST transmission consumes nearly 3.3 times as much energy as a comparable MQTT transmission [144]. This suggests that the verbose headers and synchronous nature of HTTP are not merely academic inefficiencies to be debated in software design; they generate a very literal, measurable drain on the device's physical battery chemistry. While the mathematical model assumes perfect transmission conditions, under practical deployment conditions where network congestion might cause retries, this energy gap would likely widen further. Every additional millisecond spent negotiating a secure socket translates directly into wasted Joules of finite energy.

The implications of this disparity compound significantly when extrapolated over a realistic operational timeline. For the purpose of long-term estimation, a typical use case for the "Plant Up!" sensor node can be assumed, where telemetry data is reported nominally once per hour. Over the course of a single year (comprising roughly 8,760 autonomous updates), the accumulated energy footprint expands substantially. Under an MQTT architecture, the transmission overhead alone would account for approximately 1,280 Joules per year. Under a strictly REST-based architecture, this identical workload would siphon away over 4,240 Joules annually—a difference that severely penalizes the device's operational lifespan.

This increased energy consumption naturally triggers a cascading series of secondary effects that ultimately impact long-term consumer usability. If a device utilizing REST drains its battery reserves noticeably faster than an optimally configured MQTT variant, the frequency of necessary battery replacements—or the required duration of solar recharging intervals—rises sharply. In a consumer context, frequent manual intervention entirely undermines the core premise of a "smart," autonomous sensor. A system that requires constant electrical maintenance shifts from being a convenient background utility to an active nuisance, drastically increasing the likelihood of user abandonment over time.

From an architectural standpoint, this analysis reinforces the notion that protocol efficiency holds vastly different levels of importance depending on exactly where it is ap-

plied within the system hierarchy [138]. For the backend cloud servers, the difference between parsing 312 bytes and parsing 982 bytes is structurally negligible, easily absorbed by the sheer processing capacity of modern, scalable cloud infrastructure. The server simply does not care about the extra fractions of a second it takes to parse an HTTP header. However, at the extreme edge of the network, within the stringent, physical confines of a battery-powered ESP32, these fractions of a second represent the difference between a product that lasts for several undisturbed months and one that frustrates the user by requiring a recharge every few weeks. It can therefore be reasonably concluded that optimizing protocol overhead at the device layer is perhaps the single most effective architectural intervention available for extending overall hardware longevity.

4.4.7 Conclusion

From the data gathered, it becomes evident that MQTT represents a more fitting protocol choice for the deeper sensor layer of the “Plant Up!” ecosystem. The reduction in both latency and data usage positions it favorably compared to the REST alternative. That said, REST maintains its relevance within the mobile application layer, where its familiar request-response mechanics are entirely adequate for fetching user profiles and related social content. Nevertheless, the ESP32-S3 hardware used at the edge simply cannot comfortably sustain the persistent overhead of HTTP for its high-frequency sensor reporting. Bearing these factors in mind, MQTT was ultimately selected as the standard protocol for telemetry ingestion, offering a more viable trajectory toward achieving acceptable battery longevity in a consumer-facing product.

Kapitel 5

Gamification. Gaming design patterns to keep people engaged in apps.

Author: Dennis Frenzel

5.1 Introduction

The rapid expansion of the mobile app market has resulted in an overwhelming number of new applications being launched each year, yet only a small fraction achieve long-term success. Research indicates that approximately 80–90% of apps fail within their first year, often due to low user engagement and poor retention [120, 121].

Even when initial downloads are strong, sustaining user interest remains a major challenge; studies show that up to 77% of users stop using an app within the first three days, and the majority abandon it within a month [122]. This highlights a critical issue in digital product design: attracting users is far easier than keeping them engaged.

Gamification, the use of game design elements such as rewards, progress tracking, and challenges in non-game contexts, has emerged as an effective strategy to address this problem. By leveraging intrinsic and extrinsic motivation, gamification can significantly enhance user engagement, improve retention rates, and ultimately increase the likelihood of an application's success [123].

To explore the practical application of these concepts, this thesis introduces "Plant Up!" [5], a gamified smart gardening system developed as a core component of this project. By integrating IoT sensors with a mobile application, Plant Up! is designed to transform the routine tracking of plant health into an engaging, interactive experience. It serves as a real-world case study to investigate how digital mechanics can foster sustainable user motivation and bridge the gap between initial interest and long-term engagement.

In light of these objectives, this thesis poses the following primary research question:

How can gamification be effectively applied to a gardening system to keep users more engaged in a normal activity?

5.2 Theoretical Foundations of Gamification

5.2.1 Definition of Gamification

Gamification is when game design elements and mechanics are integrated into non-game contexts. The goal of introducing these elements is to increase user engagement and motivation by making otherwise boring or routine tasks more appealing [84, 85].

However, this concept is occasionally conflated with other game-related fields, making it crucial to establish clear boundaries before diving deeper.

5.2.2 Difference between Gamification, Serious Games, and Games for Entertainment

It is important to note that gamification is not the same as serious games or games for entertainment. Although the terms sound similar, they describe different concepts. Gamification refers to the integration of game elements into non-game contexts in order to increase user engagement and motivation. Serious games are complete games that are explicitly designed for a primary purpose other than entertainment, such as education, training, or simulation. For example, a flight simulator used for pilot training or a surgery simulation used for medical training. Games for entertainment, on the other hand, are produced only for enjoyment, without any educational or functional objective [86].

5.2.3 Why Gamification Works in Non-Game Contexts

Understanding what gamification is and what it is not naturally leads to exploring its efficacy. While many people question the effectiveness of gamification, research has shown that it can be an effective strategy for enhancing user engagement and motivation. Specifically, “Many older generations associate the word ‘game’ with ‘wasting time,’” says Dr. Donna Davis, associate professor and director of the Strategic Communication Master’s Program at the University of Oregon [124]. Despite this skepticism, data demonstrates that gamification can still bridge these generational gaps when properly implemented. This raises the questions: Why and how?

According to the Haufe Akademie [87], the effectiveness of gamification lies deep within the human psyche. It addresses the basic needs of autonomy, competence, and social relatedness, which form the foundation of human motivation. How gamification achieves this can be explained using three central mechanisms.

- **1:** Gamification integrates the human need for autonomy, competence, and social relatedness not merely through external rewards, but by actively supporting these needs.
- **2:** It redefines user interaction with a system. Instead of removing all difficult or boring tasks to maximize efficiency, gamification introduces voluntary challenges that encourage people to improve themselves. These tasks are transformed into engaging activities in which progress is clearly visualized and supported by immediate feedback mechanisms such as points and badges.
- **3:** Gamification uses social interaction to build a sense of community. Features such as cooperative challenges and leaderboards satisfy the desire to be noticed by others, increasing attachment to the activity through competitive or collaborative motivation [87].

As these mechanisms highlight, the core driving force behind any successful gamified system is human motivation. To fully grasp how these game elements interact with user behavior, we must delve deeper into the underlying psychological theories.

5.2.4 Motivation Theory

To understand why gamification can be effective in practice, it is necessary to examine the foundations of human motivation. Motivation is the psychological force that initiates, guides, and sustains goal-directed behavior. In the context of gamification, a key distinction is made between engaging in an activity because it is inherently rewarding (intrinsic motivation) and engaging in it to obtain an external outcome (extrinsic motivation) [88].

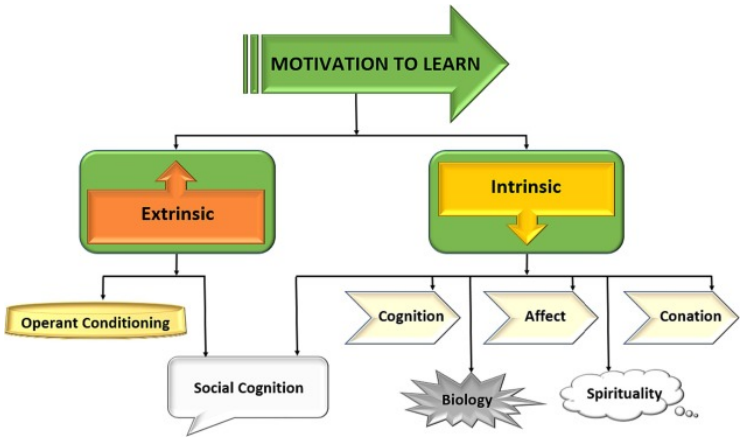

Abbildung 5.1: Classification of Motivation Types [88].

5.2.4.1 Intrinsic Motivation (The Engine of Engagement)

Intrinsic motivation is when a person engages in an activity because it is satisfying, enjoyable, or meaningful, while it also comes from within the person and not from external factors [88]. According to Self-Determination Theory (SDT) [89], intrinsic motivation is

supported when three basic psychological needs are fulfilled: Autonomy, Competence, and Relatedness.

When these needs are satisfied, individuals demonstrate increased curiosity, creativity, and persistence [88]. But first, let's define what Self-Determination Theory is.

Self-Determination Theory (SDT) in Gamification
Self-Determination Theory establishes a framework for human motivation based on three fundamental psychological needs: autonomy, competence, and relatedness. Meeting these needs fosters self-determined motivation, which drives improved performance, persistence, and creativity. Failing to support them can significantly reduce motivation and well-being [88, 89]. Each of these three needs plays a specialized role when applied to a gamified context, starting with autonomy.

Supporting Autonomy
Autonomy refers to the user's sense of control over their actions and decisions. This is achievable with integrating optional tasks, multiple paths to success, and customization options [88].

While autonomy provides users with the freedom to choose, this freedom must be paired with clear feedback on their progress. This leads to the second psychological need: competence.

Supporting Competence
Competence refers to the user's sense of accomplishment and skill development. It is mostly done by integrating experience points, progress indicators, adaptive difficulty, and immediate feedback loops [88].

However, even when users feel autonomous and competent, human beings remain inherently social creatures. Gamification systems capitalize on this through the final pillar of SDT: relatedness.

Supporting Relatedness
Users are more engaged when they feel socially connected, hence integrating social feeds, cooperative goals, peer comparison, and mentorship features is a go-to strategy [88].

Fulfilling these three psychological needs (autonomy, competence, and relatedness) builds a strong foundation for intrinsic motivation. In practical gamification, however, satisfying these inherently internal needs often requires the careful distribution of external rewards.

Rewards that Support Intrinsic Motivation
Motivators that support intrinsic motivation are linked to a personal desire to master, explore, and enjoy an activity. They are the main motivator for long-term commitment, when implemented right, trying to keep them as informational feedback rather than controlling mechanisms is the key to succeeding [90] Most of the time, just like in Plant Up! [5], those are accomplished by integrating, badges, that indicate achievement and mastery, experience points, virtual currency for customization and titles [90].

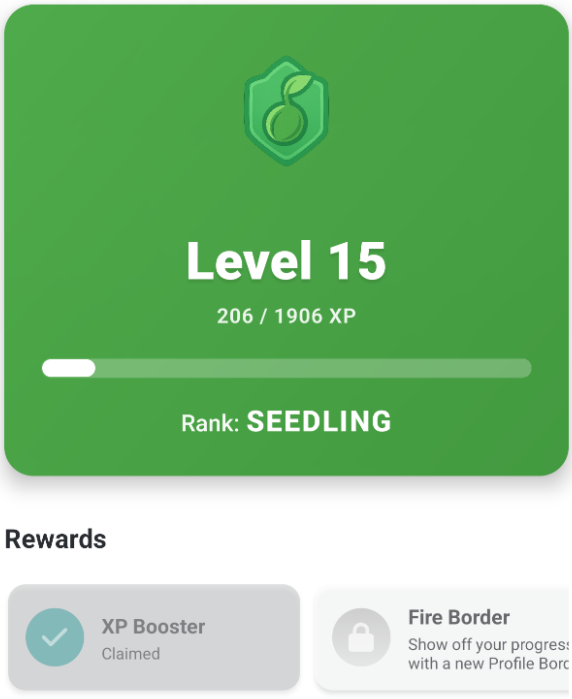

Abbildung 5.2: Level System and Rewards in Plant Up!.

While well-designed rewards enhance intrinsic motivation, their improper use can quickly have the exact opposite effect.

Rewards that Undermine Intrinsic Motivation
Some rewards can be counterproductive when they seem controlling or worthless [90]. Such as over-rewarding insignificant actions, excessive penalties that induce stress, rankings without fairness or relevance, and when rewards replace genuine interest in the activity itself [90]. This often makes those rewards feel like a chore, just like the Duolingo Streak [94].

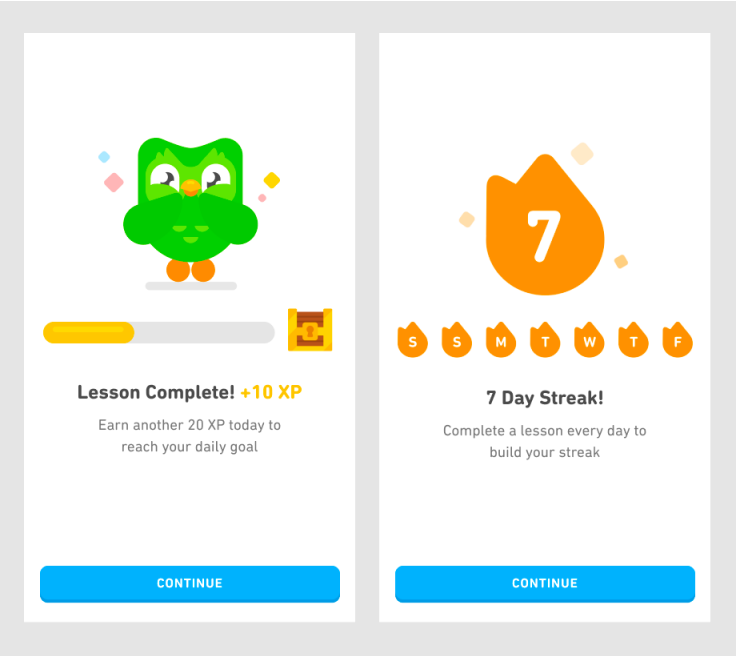

Abbildung 5.3: Duolingo Streak [94].

5.2.4.2 Extrinsic Motivation (The Spark)

Extrinsic motivation involves performing an activity in order to receive a reward or avoid negative consequences. Common gamification elements such as points, badges, and leaderboards are examples of extrinsic motivators [88].

- **Controlling Extrinsic Motivation:** When rewards are perceived as chore-like, they can reduce intrinsic motivation, resulting in short-term compliance but reduced long-term engagement. A typical example is a streak-based system that pressures users to perform regularly, which can lead to stress and burnout.[88].
- **Autonomy-Supportive Extrinsic Motivation:** When rewards provide informational feedback or signal competence, they can be internalized and reinforce intrinsic motivation. Feedback such as positive reinforcement for progress or improvement is more effective than obligation-based incentives.[88].

Effective gamification strategies use extrinsic rewards not as the primary objective, but as a means of supporting intrinsic motivation [87, 88]. Ultimately, the goal is to strike a delicate balance between providing immediate incentives and preserving the user's inherent interest.

The Role of Extrinsic Rewards
Gamification is not merely about adding rewards, but about how they are implemented. Extrinsic motivators can be powerful drivers of engagement but may reduce creativity and encourage mechanical behavior. This can result in retention driven by obligation rather than enjoyment [90]. When used carefully, extrinsic motivators can still provide

short-term motivational boosts that foster intrinsic interests such as mastery or competition [90, 88].

5.2.5 Attribution Theory

While intrinsic and extrinsic motivations explain the *why* of user engagement, how users perceive their own success or failure within these systems is equally critical. This perception is the focus of Attribution Theory.

Attribution theory describes how individuals explain success and failure. In gamified systems, it is important that users get rewarded for their own effort and strategies rather than luck or fixed ability. Implementing effort-based feedback can increase motivation and encourage continued user engagement. Additionally, designing penalties and failure in such a way that it only seems temporary and improvable can lead to less frustration, hence making the user more likely to continue using the app [92].

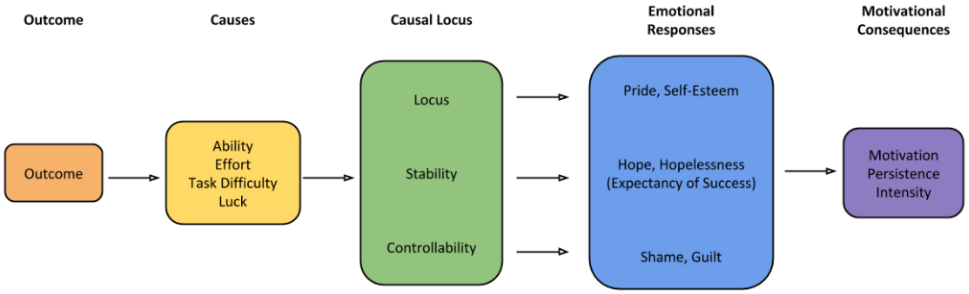

Abbildung 5.4: Attributional Process of Motivation.

Closely related to how users attribute their success or failure is their initial assessment of a task before even attempting it. Expectancy-value theory sheds light on this preparatory phase of motivation.

5.2.6 Expectancy-Value Theory

Expectancy-value theory explains what motivates a user before engaging in an activity: the expectation of success and the subjective value of the task. When a user thinks that they can succeed in an activity and values it, they are more likely to engage in it.

1. **Expectancy for Success:** Expectancy for success can be defined by the question: “How likely is it that I will succeed in this activity?” In games and gamified systems, this belief is answered with a clear definition of goals, transparent rules, the difficulty of the task, and feedback provided during the activity. To make the user have a clear vision of their success, it is important to provide clear objectives, transparent mechanics, and visible indicators of progress. When the criteria for the question “Is this challenge achievable?” are met, the user is more likely to invest effort in it [93].

2. **Subjective Task Value:** Subjective Task Value describes whether the task is worth the effort, which varies for each user. This value consists of several components, including intrinsic value, the importance of performing well, usefulness for future goals, and effort, such as time investment. In games, a value can be increased by making the user progress faster, reducing frustration, or making those tasks relevant for future goals [93].

Ultimately, motivation theories aim not just for temporary spikes in activity, but for sustained user participation. Translating these theoretical concepts into lasting behaviors requires effective engagement mechanics.

5.2.7 Engagement and Habit Formation

5.2.7.1 Feedback Loops

A feedback loop is an important component of gamification, as it provides users with immediate feedback on their progress and achievements.

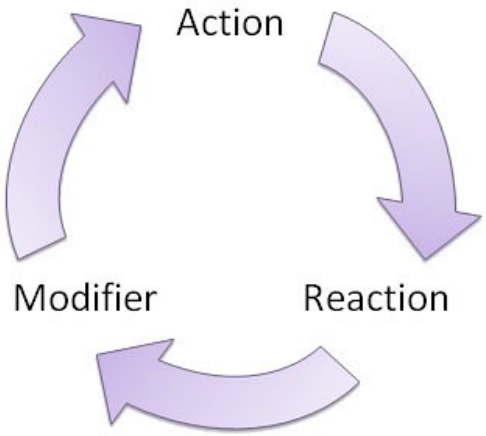

Abbildung 5.5: Feedback Loop [96].

It is important to note that the feedback loop should consist of both positive and negative feedback.

Most people prefer to receive positive feedback, such as badges, experience points, virtual currency, and titles. However, it is important to note that negative feedback can also be a motivator. Not only can negative feedback, such as penalties, warnings, or punishments, be a wake-up call for users to start improving their behavior, but it can also motivate them to avoid those penalties in the future [97].

5.3 Gamification Design Patterns

Building upon the theoretical foundations of motivation, gamification design patterns offer concrete ways to translate those psychological needs into actual application features.

5.3.1 Core Game Mechanics

When looking at basic mechanics like XP and badges, it helps to understand the psychology behind them. According to Yu-kai Chou's Octalysis framework, these elements are driven by what he calls Core Drive 2: Development & Accomplishment. This drive taps into our internal desire to make progress, develop skills, and overcome challenges [125]. For example, in a gardening application, the mundane task of watering a plant becomes more engaging when that action yields immediate visual feedback, such as earning points that demonstrate an expanding "green thumb".

However, Chou points out that points and badges alone are just "Feedback Mechanics", as they serve as milestones to show users they are getting closer to a meaningful "Win State." If a system lacks a balanced progression curve, these features quickly fade, feeling hollow because they no longer satisfy the human need for genuine competence. In the context of cultivating plants, this means ensuring that leveling up reflects real-world mastery over time, rather than just endlessly repeatedly tapping a button. On the other hand, adding components like virtual currency engages another motivator: Core Drive 4: Ownership & Possession [125].

As seen in the Plant Up! application (see Figure 5.6), users earn virtual currency which they can spend in the in-game shop on rewards like XP Boosts or Streak Freezes. When users earn currency, they feel compelled to hold onto it, improve it, and protect their stash [5].

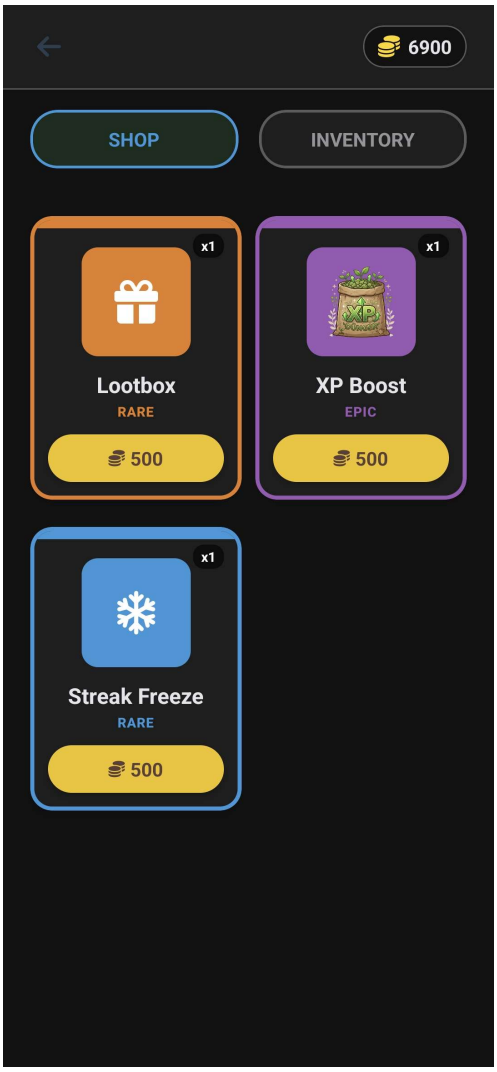

Abbildung 5.6: In-Game Shop and Virtual Currency in Plant Up!

In the end, these core mechanics act as the structural skeleton of an app, ensuring that the time a user spends feels both tangible and organized.

However, while core mechanics establish the baseline of a gamified experience, they alone are often not enough to retain users over the long term. This is where more complex and urgent techniques come into play.

5.3.2 Advanced Engagement Patterns

To hook users on a deeper level, developers often turn to what Chou labels “Black Hat” motivation, specifically Core Drive 8: Loss & Avoidance and Core Drive 6: Scarcity & Impatience [125]. A prime example of this is the concept of daily streaks. Streaks build a powerful “Sunk Cost” effect: users become less motivated by the actual reward and more driven by the fear of losing the progress they’ve already stockpiled. Applied to gardening, this effectively creates a “virtuous cycle” where skipping a day not only breaks a digital streak, but could also tangibly lead to letting a beloved plant wither.

Similarly, time-based rewards and rarity mechanics lean heavily on scarcity. An action or item feels inherently much more valuable simply because it isn't always available to everyone. In a plant care system, this can be mapped to seasonal events or blooming periods, where certain badges or virtual items are only available during a rare, time-sensitive harvest. While these "Appointment Dynamics" are undeniably effective for short-term retention, they walk a fine line. Relying too much on Black Hat drives can make users feel like slaves to the system, creating a sense of obsession rather than genuine fun [125]. The trick is to use these mechanics to inject just enough urgency, such as an alert to water a struggling succulent, without burning the user out.

Beyond individual motivation and time-sensitive triggers, another massive driver for sustained participation lies in human interaction. This leads to an entirely different set of rules focusing on social dynamics.

5.3.3 Social Gamification

Nothing drives human behavior quite like the desire to relate to, and occasionally surpass, our peers. Social mechanics are heavily fueled by Core Drive 5: Social Influence & Relatedness [125]. In a gardening ecosystem, this could be achieved by allowing users to share pictures of their thriving monstera plant or showcasing their garden's overall health score. Leaderboards are the classic "Social Comparison" tool here. They can spark healthy competition, but if designed poorly, they backfire completely. If the points gap between the top player with a massive greenhouse and a newcomer with just one cactus is impossibly huge, it often leads to "Antisocial Envy," causing the new user to just give up.

To balance this out, it's crucial to implement cooperative designs alongside competitive ones. Group quests, for instance, utilize "Social Proof" to build community, such as challenging a group of friends to collectively log twenty days of perfect plant care. When users feel pressure to contribute to a collective goal, they are much more likely to stay engaged. Taking this a step further, adding mentorship or recognition features helps transition top players from chasing individual achievements to earning high-level status within the community by offering guidance to novice plant owners [125]. When competition and cooperation are properly balanced, the system satisfies both our need to stand out and our need to belong.

While combining all these mechanics, such as core progression, urgency, and social interaction, can create highly engaging systems, they must be implemented with care. If pushed too far, gamification can quickly become manipulative, highlighting the need for firm ethical boundaries.

5.3.4 Failure States & Ethical Design

A gamification system fails when it relies almost exclusively on Black Hat drivers. Over time, this leads to heavy burnout, leaving users resentful [125]. For instance, harassing a user with aggressive penalties and notifications every time their soil moisture drops

slightly could make the hobby of gardening feel like a stressful chore rather than a relaxing activity. Chou specifically warns against “Function-Focused Design”. This happens when a designer stops caring about the human experience and instead optimizes purely for metrics like clicks, daily active users, or time-on-site.

Another pitfall is the “Over-Justification Effect.” If a system gives out too many extrinsic rewards (like endless virtual currency for every single drop of water given to a plant), it eventually “kills” the user’s genuine intrinsic interest in the task. Even worse, dark patterns can emerge when loss aversion and scarcity are weaponized, manipulating people into spending real money or time they never intended to part with. To design ethically, there has to be a pivot toward “White Hat” drivers, which are elements that give the user a sense of meaning and empowerment. When a session ends, the user should feel a real sense of accomplishment for having kept their plants alive and thriving, satisfied and in control, not manipulated [125].

5.4 Why Gamification Works for Normal Activities

Applying game design elements to everyday situations has grown from a simple marketing trick into a powerful way to encourage positive habits [126]. When it comes to a very grounded but essential activity like gardening, the biggest hurdle is usually the gap between the effort you put in today and the results you might only see weeks later. This section breaks down exactly why gamification is such an effective strategy for keeping users engaged with a gardening system over the long haul. It specifically focuses on how adding a digital layer can actually enhance the physical experience of taking care of plants. By looking at the psychological factors that drive motivation, the following subsections will explore how repetitive chores are transformed, how immediate feedback bridges the gap between slow nature and fast technology, and how emotional ownership naturally keeps users coming back.

5.4.1 Transformation of Mundane Tasks

Gardening is often seen as a relaxing and therapeutic hobby, but it still involves a lot of repetitive, manual chores like watering, weeding, and checking the soil. For a lot of people, especially those who are used to the fast pace of modern digital life, these tasks can start to feel like a tedious obligation after a while. Gamification tackles this problem head-on by layering a structured sense of progress over these basic physical actions. As Deterding et al. define it, gamification is simply taking elements we love in games and applying them to non-game situations to make the experience more engaging [133]. In a gardening app, this usually takes the form of points, leveling systems, and daily streaks.

The magic of these design elements is that they make invisible progress visible. You might not see a plant physically grow any taller right after you water it [134], but a digital interface can instantly give you experience points (XP) for doing the right thing. This immediate feedback is incredibly important for balancing out different types of motivation. According to Self-Determination Theory, intrinsic motivation means doing something

just because you enjoy it, while extrinsic motivation means doing it for a specific external reward [135]. By handing out points and badges, the system provides a fun, extrinsic push that keeps the user consistent until the real, intrinsic reward, like seeing a new leaf or a blooming flower, finally reveals itself.

On top of the points system, there is the concept of streaks, which has become incredibly popular in modern apps because it taps directly into our psychological need to be consistent. When someone realizes they have built up a “10-day watering streak,” they are far less likely to skip a day. The fear of losing that clean track record suddenly becomes a stronger motivator than the laziness of skipping a chore. This idea fits perfectly with the Fogg Behavior Model, which explains that a behavior only happens when motivation, ability, and a clear prompt all line up at the exact same time [136]. In this case, the gamified app acts as that persistent, friendly prompt. By slicing a massive, months-long goal like a successful harvest into tiny, daily digital milestones, the system turns a slow and “normal” activity into a fun series of rewarding micro-tasks.

Once these mundane chores are broken down into manageable digital steps, the next challenge is ensuring that users confidently know their efforts are working. This leads directly to the importance of real-time confirmation.

5.4.2 Feedback in Real-World Processes

One of the biggest hurdles to sticking with gardening is simply the biological latency of plants. Unlike clicking a button on a website and getting an instant result, natural growth takes time. A seed might take weeks to sprout, and a plant can take months before it finally flowers [134]. This massive disconnect between our modern expectation for instant speed and the slow, quiet pace of nature is often the exact point where beginners lose interest and drop off. Gamification bridges this specific gap by bringing instant feedback loops into the real world, often powered by smart sensors. For example, the moment a sensor detects that the soil moisture has hit the perfect level after watering, the app can flash a glowing green icon, play a joyful sound, or fill up a progress bar.

This creates a tight, closed feedback loop, which is a cornerstone of behavioral psychology. Skinner’s theory of operant conditioning suggests that any behavior followed by a pleasant consequence is highly likely to be repeated [132]. In a smart gardening app, the sensor data acts as the missing link between physical effort and digital celebration. When a user gets a push notification congratulating them that their “Plant Health Index” has improved, it triggers a tiny dopamine hit in the brain. This reinforces the daily habit of checking in on the system. This instant feedback is an absolute game-changer for beginner gardeners who often worry if they are watering too much or too little. The digital system acts as a supportive coach, providing immediate reassurance that their physical actions were correct.

Taking this a step further, beautiful data visualization heavily amplifies this empowering effect. Instead of staring at what looks like a static pot of dirt, the user can pull up a dynamic dashboard displaying growth curves, current nutrient levels, and estimated harvest dates. Being able to visualize this progress satisfies our deep-seated need for

information and a sense of control over our environment [128]. In fact, industry reports consistently show that users are far more likely to stick with a platform if it offers them highly personalized, constantly updating data [130]. By turning a slow biological process into a lively digital dashboard, the system completely removes the frustration of waiting. Instead, it replaces it with the satisfaction of managing a responsive, living entity.

HIER KOMMT EIN BILD REIN VON PLANT UP! UND DIE STATISTIKEN!

When users feel capable and informed, they naturally start to care more about the entity they are nurturing, which brings us to the most powerful long-term motivator of all: emotional connection.

5.4.3 Emotional Attachment & Ownership

While points and instant feedback are fantastic for getting people started, truly long-term engagement is anchored by the emotional connection a user forms with the system. Gamification can heavily foster this bond through customization and by giving a physical plant a distinct digital identity. Simply allowing a user to give their plant a nickname or assign it a cute avatar instantly shifts the app from being a sterile tool into feeling like a companion. This deep sense of ownership is rooted in what behavioral economists call the “IKEA effect.” It is a cognitive bias where people place a significantly higher personal value on things they have helped build or actively labored over themselves [127]. If a user spends three months carefully logging, watering, and nurturing a specific plant within the app, they build up a level of personal investment that makes it incredibly hard to just walk away and abandon the project.

HIER KOMMT EIN BILD REIN VON PLANT UP! UND DER QUESTS/SKINS/EIGENE DIGITALE PLANT/ SPRIG MASKOTT!

This exact type of investment is the final core component of Nir Eyal’s famous Hook Model, which outlines a four-phase process for creating habit-forming products. The cycle consists of a trigger, an action, a variable reward, and finally, an investment [131, 129]. Within a smart gardening system, the “investment” happens every single time the user inputs a new piece of data, unlocks a milestone, or adjusts their garden layout. The more time and effort the user pours into customizing their virtual space and tracking their physical plant, the more irreplaceable the system becomes to them. All of that collected historical data acts as an emotional digital memory of their hard work, constantly validating their identity as a capable and successful gardener.

Ultimately, gamification cements this habit by wrapping the user’s daily actions into an overarching personal narrative. Being able to look back at years of growth allows users to vividly see how much their own skills have improved, effectively elevating a casual hobby into a satisfying journey of mastery. It is well documented that apps leveraging personalization and emotional hooks boast much higher retention rates than those offering only raw data and functional menus. In the context of a gamified garden, the user is not just keeping a plant alive. They are tangibly “leveling up” their real-world environment and their own capabilities. It is this powerful mix of emotional attachment, earned

ownership, and personal growth that guarantees a user will stay committed long after the initial novelty of the app has worn off. By weaving these gamification principles together, a completely normal gardening routine is successfully transformed into a deeply engaging, habit-forming experience that gracefully unites digital rewards with beautiful, real-world results.

5.5 Case Study: Gamification in Plant Up!

“Plant Up!” [5] is a gamified smart gardening system that combines IoT sensors with a mobile application to provide users with a fun and engaging way to monitor and care for their plants. How does the gamification concept in “Plant Up!” [5] work?

5.5.1 Evaluation and Discussion

The core goal of integrating these gamified mechanics into the Plant Up! [5] system is to completely bridge the gap between dry, technical plant monitoring and vibrant, long-term user engagement.

5.5.1.1 Meaningful Habit Formation

Looking closely at core mechanics like XP, daily streaks, and the in-game “XP Booster,” it becomes clear that the system is heavily leveraging classical behavioral psychology to foster sustainable habit formation. However, what sets Plant Up! [5] apart is that these rewards are not just arbitrary digital tasks. Because they are directly tied to real-time data pulled from smart IoT sensors, the rewards are grounded in the actual biological needs of the plants. This creates a framework of genuine, meaningful gamification. By providing structured extrinsic rewards (XP and coins) during the slow biological latency phases, the system successfully bridges the motivation gap until the intrinsic reward (e.g., a healthy, blooming plant) is achieved. This effectively puts the core concepts of Self-Determination Theory (SDT) [135] into practical application.

This smooth transition from extrinsic nudges to intrinsic satisfaction naturally leads into how the system supports beginners who have yet to develop those intrinsic skills.

5.5.1.2 Lowering the Entry Barrier

One of the primary strengths of the Plant Up! [5] approach is how effectively it lowers the barrier to entry for complete beginners. This serves as a significant structural advantage because it directly mitigates the steep learning curve often associated with botany, actively supporting the “Competence” pillar of SDT. Through the use of “Sprig,” the friendly NPC guide, and a system of structured quests, the application translates complex botanical sensor data into simple, actionable feedback (see Figure 5.7).

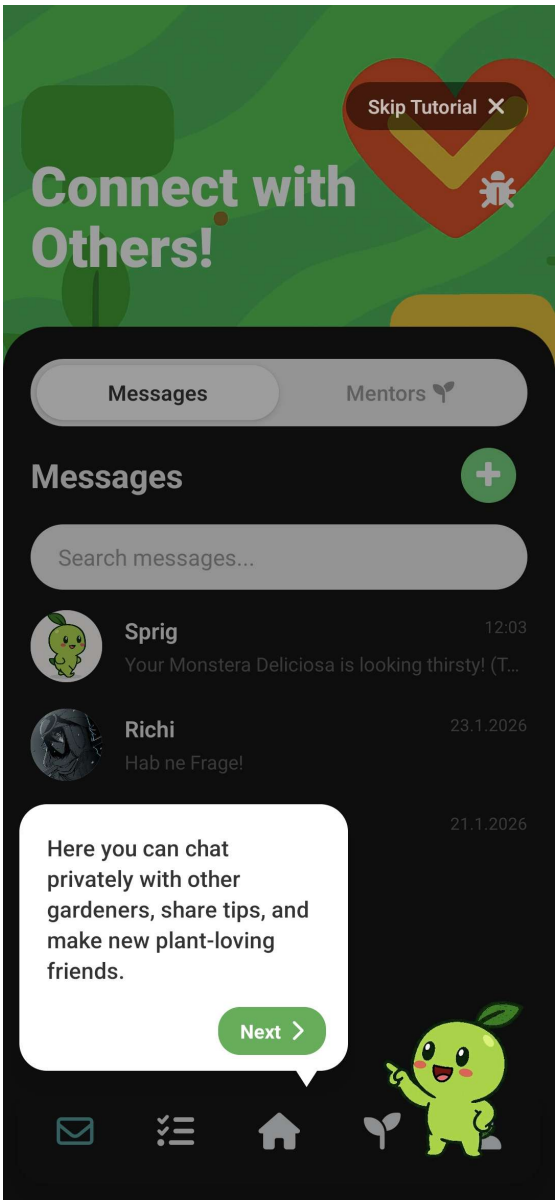

Abbildung 5.7: The Plant Up! NPC guide “Sprig” providing tutorials and easily understandable feedback to help new users [5].

These elements directly address a massive pain point in the gardening hobby: the feeling of being completely overwhelmed by plant care demands. Furthermore, the inclusion of items like the “Freeze Streak” and “XP Boosters” in the shop demonstrates a sophisticated understanding of how to retain players over the long haul. These specific features actively mitigate the heavy “loss aversion” a user feels when they inevitably miss a day of care. By offering a safety net, the system prevents the devastating loss of progress that usually causes users to abandon an app entirely in a moment of frustration.

While the inclusion of such mechanics perfectly addresses beginner anxiety and builds early engagement, it introduces a separate set of risks that must be managed as the user matures.

5.5.1.3 Potential Long-Term Challenges

Relying heavily on extrinsic motivators like virtual coins and titles in the Plant Up! [5] ecosystem does present some potential long-term challenges. This over-reliance represents a structural disadvantage because it can condition users to engage only when an external reward is present, rather than developing a genuine, self-sustained interest in the activity [135]. While these mechanics are incredibly effective for building the initial habit, there is a lingering risk of the "over-justification effect." This happens when users eventually start prioritizing the digital rewards and leaderboards over the actual, intrinsic health of their physical plants. If the gamification layer becomes too dominant, the primary utility of the IoT sensor data, which is ensuring plant survival, could be completely overshadowed by the pursuit of online status. Additionally, designers must be wary of "statistical anxiety." This can easily arise from public social features if users begin to feel intense pressure to maintain perfect streaks, potentially turning what should be a relaxing hobby into an unexpected source of digital stress [5].

To counteract these antisocial or stressful effects, the system relies heavily on communal features that foster the "Relatedness" aspect of SDT, requiring a robust technological foundation.

5.5.1.4 Technical Architecture & Community

From a technical perspective, the overall architecture of Plant Up! [5] efficiently handles the constant stream of sensor and user-generated content, such as shared plant-care videos and social posts, by utilizing an ASP.NET API coupled with Supabase. This social dimension is absolutely crucial for the long-term viability of the project because it elevates the app from being just a solo monitoring tool into a massive, knowledge-sharing community. By fulfilling the human need for social connection, it actively supports the third pillar of SDT (Relatedness).

Looking ahead, future technical enhancements will likely need to focus heavily on aspects like Android animation normalization. Ensuring that the gamification UI remains perfectly fluid across a massive variety of different mobile hardware is essential, as any technical friction or lag in the animations can instantly break the "flow" state required for an immersive gaming experience [5].

5.6 Evaluation & Discussion

5.6.1 Expected Engagement Improvements

- **Increased daily active users:** Gamification features like streaks are expected to drive daily logins.
- **Longer session duration:** Interactive elements and progression systems encou-

rage users to spend more time in the app.

- **Better plant care consistency:** Reminder quests and rewards aim to regularize care activities, leading to healthier plants.

5.6.2 Risks & Limitations

- **Users gaming the system:** Users might perform unnecessary actions just to earn points (e.g., over-watering).
- **Motivation drop after novelty:** The engagement spike from new features may fade ("novelty effect").
- **Balance issues:** Ensuring rewards feel earned but attainable is difficult to fine-tune.

5.6.3 Comparison to Non-Gamified Apps

- **Traditional gardening apps:** Focus solely on information and scheduling.
- **Why they lose users:** Lack of immediate feedback and emotional hook leads to abandonment when the initial enthusiasm fades or the task becomes a chore.

Abbildungsverzeichnis

AI-Generated Figures

The following figures were generated using an AI image generation tool. The prompts used are listed below for transparency.

Figure 2.3 *Cross-section of a plant pot showing non-uniform soil moisture distribution. One side is visibly wetter (darker soil) near the surface where water was poured, while the opposite side remains dry (lighter soil). A soil moisture sensor probe is inserted on the dry side, illustrating how a single-point measurement can be misleading. Clean diagram style, white background, labeled arrows indicating wet zone, dry zone, and sensor position.*

Figure 2.5 *Professional flowchart for an academic thesis on white background with minimal style. Starting node: Sensor Data Received. Process box: Read Values (Soil Moisture, Temperature, Humidity, Light). Diamond decision: Compare against Thresholds. Three output paths: Green Zone (optimal, no action required), Yellow Zone (suboptimal, attention recommended), Red Zone (critical, immediate action required). All paths merge into: Generate Care Recommendation, then Display to User via App. Standard flowchart shapes, sans-serif font, colored zone boxes (green, yellow, red).*

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